Repaso de los conceptos de metaprogramación

En donde repasamos los conceptos que vimos y consideramos características deseables para un sistema reflexivo.

Universidad Nacional de Quilmes

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CONCEPTS AND EXPERIMENTS IN COMPUTATIONAL REFLECTION

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ABSTRACT. This paper brings some perspective to various concepts in computational reflection. A definition of computational reflection is presented, the importance of computational reflection is discussed and the architecture of languages that support reflection is studied. Further, this paper presents a survey of some experiments in reflection which have been performed. Examples of existing procedural, logic-based and rule-based languages with an architecture for reflection are briefly presented. The main part of the paper describes an original experiment to introduce a reflective architecture in an object-oriented language. It stresses the contributions of this language to the field of object-oriented [illegible] illustrates the new programming [illegible] examples show [illegible]

Reflective architectures have already been realised for procedure-based (Smith,1982), logic-based (Weyhrauch, 1980) and rule-based languages (Davis,1982). The second part of the paper (sections 6 to 8) discusses the realisation of a reflective architecture in an object-oriented language (further on called OOL). Existing OOLs only support limited, ad-hoc reflective facilities, which leads to limitations and unclear designs, and consequently to problems in [illegible] However, over the years OOL [illegible] providing more [illegible]

Gilad Bracha

David Ungar

Mirrors: Design Principles for Meta-level Facilities of Object-Oriented Programming Languages

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ABSTRACT

We identify three design principles for reflection and metaprogramming facilities in object oriented programming languages. *Encapsulation:* meta-level facilities must encapsulate their implementation. *Stratification:* meta-level facilities must be separated from base-level functionality. *Ontological correspondence:* the ontology of meta-level facilities should correspond to the ontology of the language they manipulate. Traditional/mainstream reflective architectures do not follow these precepts. In contrast, reflective APIs built around the concept of *mirrors* are characterized by adherence to these three principles. Consequently, mirror-based architectures have significant advantages with respect to distribution, deployment and general purpose metaprogramming.

Categories and Sub

D.3.2 [

```
Car anotherCar = theCarsClass.newInstance();
Class theCarsSuperclass = theCarsClass.getSuperclass();
```

Looking at the APIs of such a system, we expect to see something like:

```
class Object {
Class getClass();
...
}
class Class {
Class getSuperclass();
// many other methods: getMeth
}
```

Definición de comportamiento para objetos específicos

En donde revisamos el metamodelo de Ruby y vemos qué es una *singleton class*.

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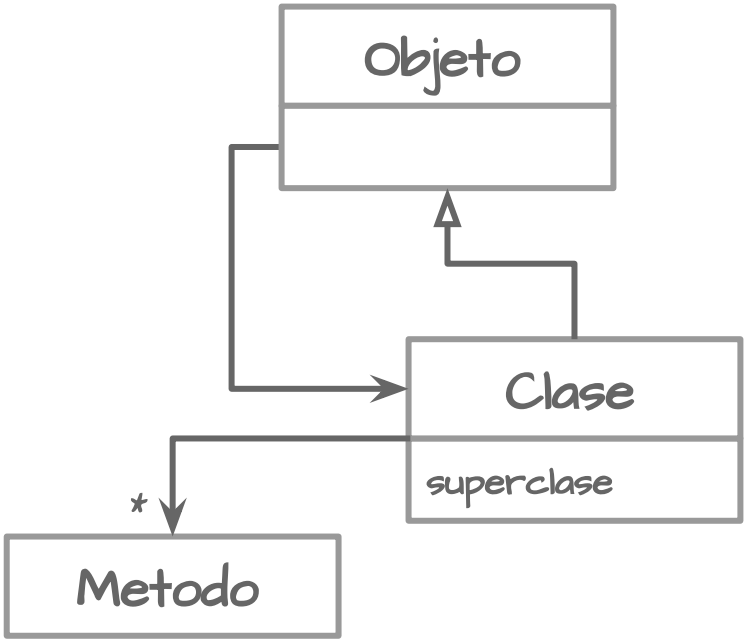
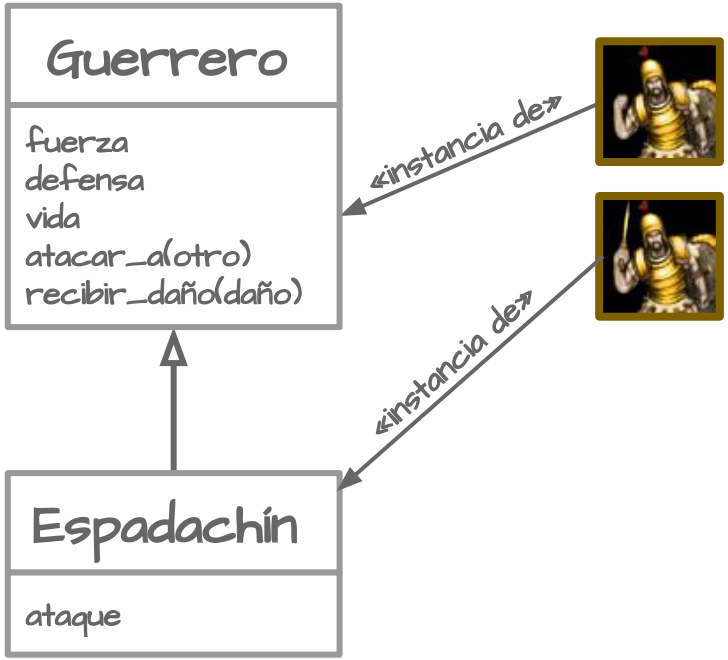
Repaso

Modelos y metamodelos

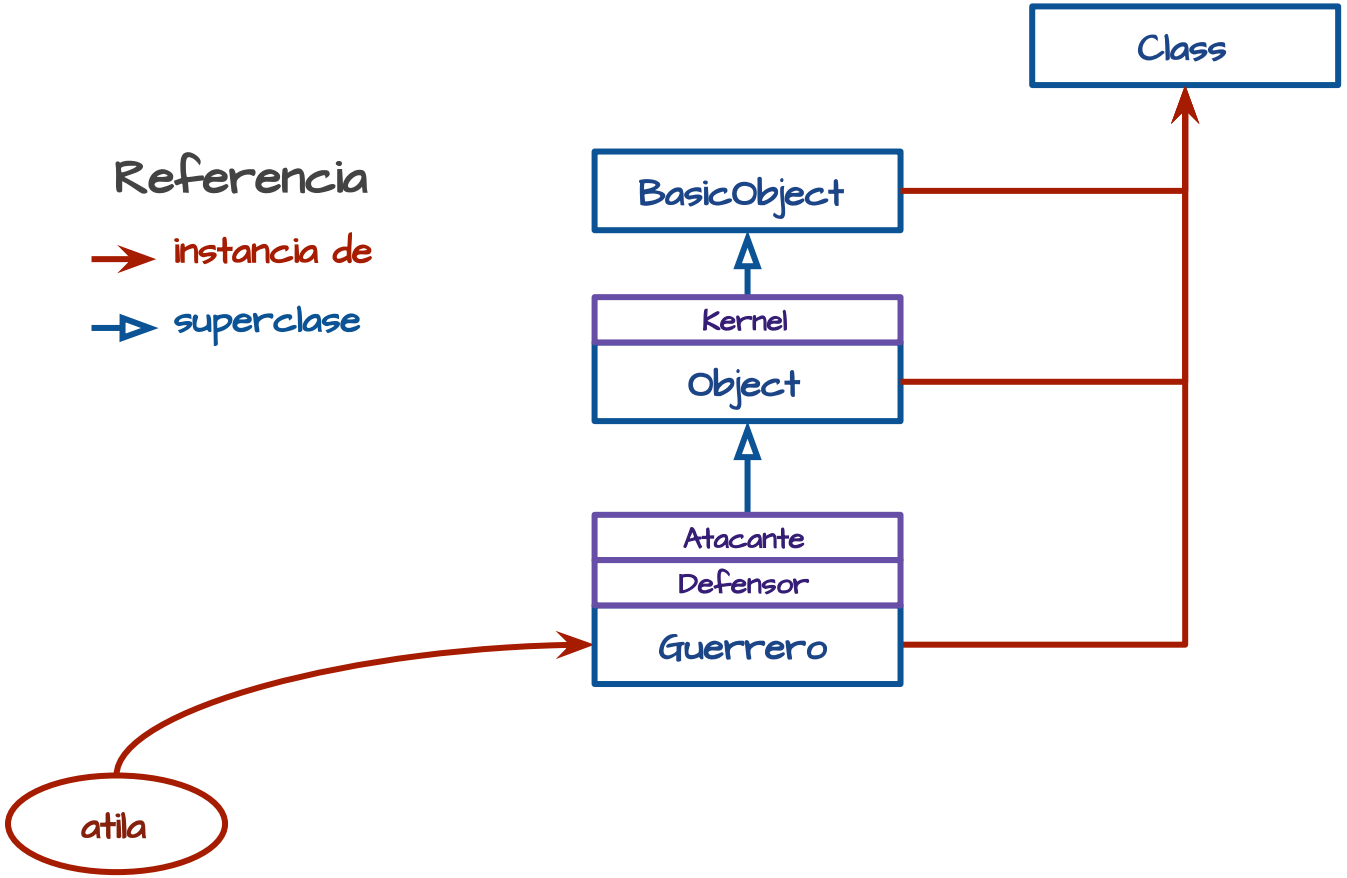
Metamodelo



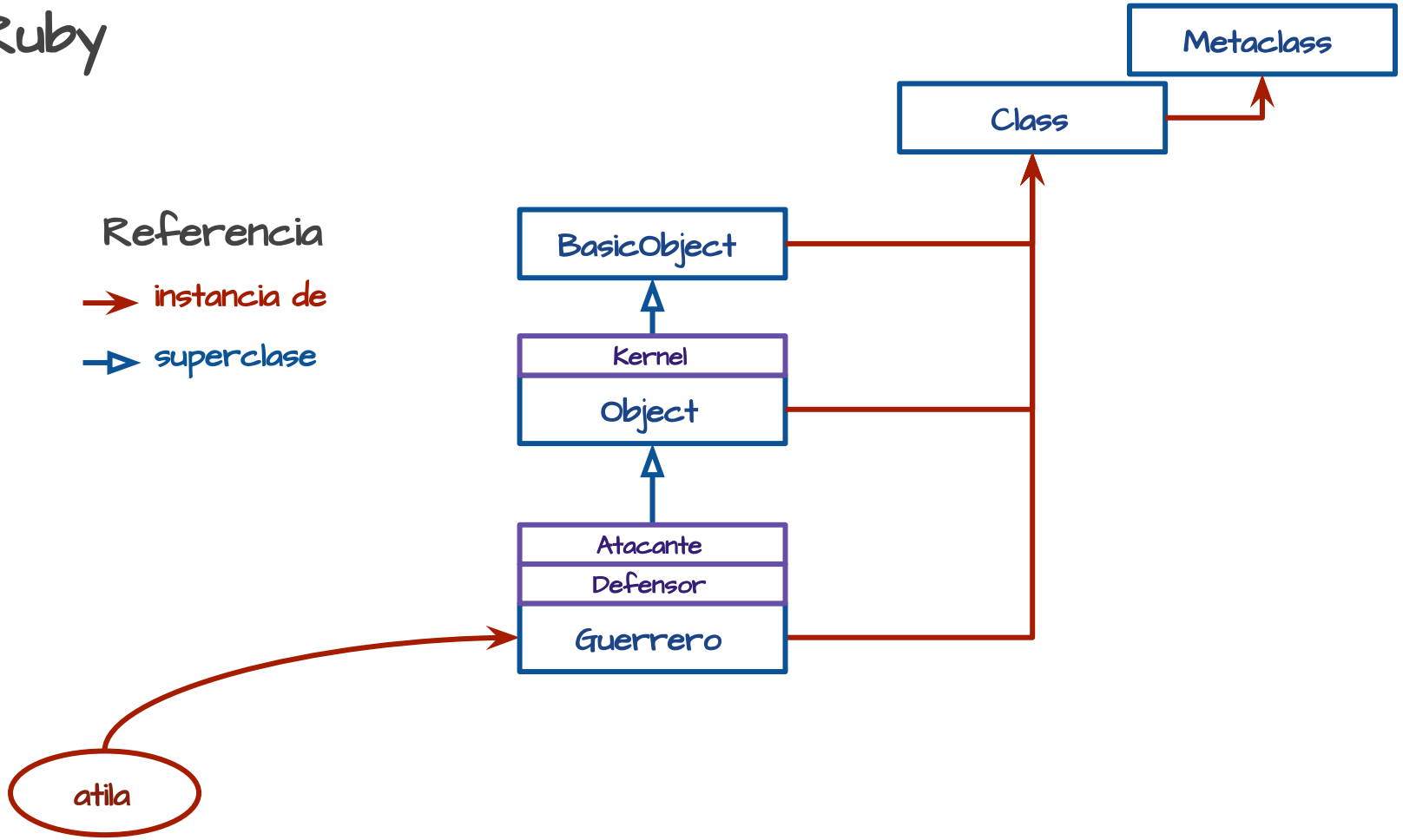
Modelo



Parte del metamodelo de Ruby



Parte del metamodelo de Ruby



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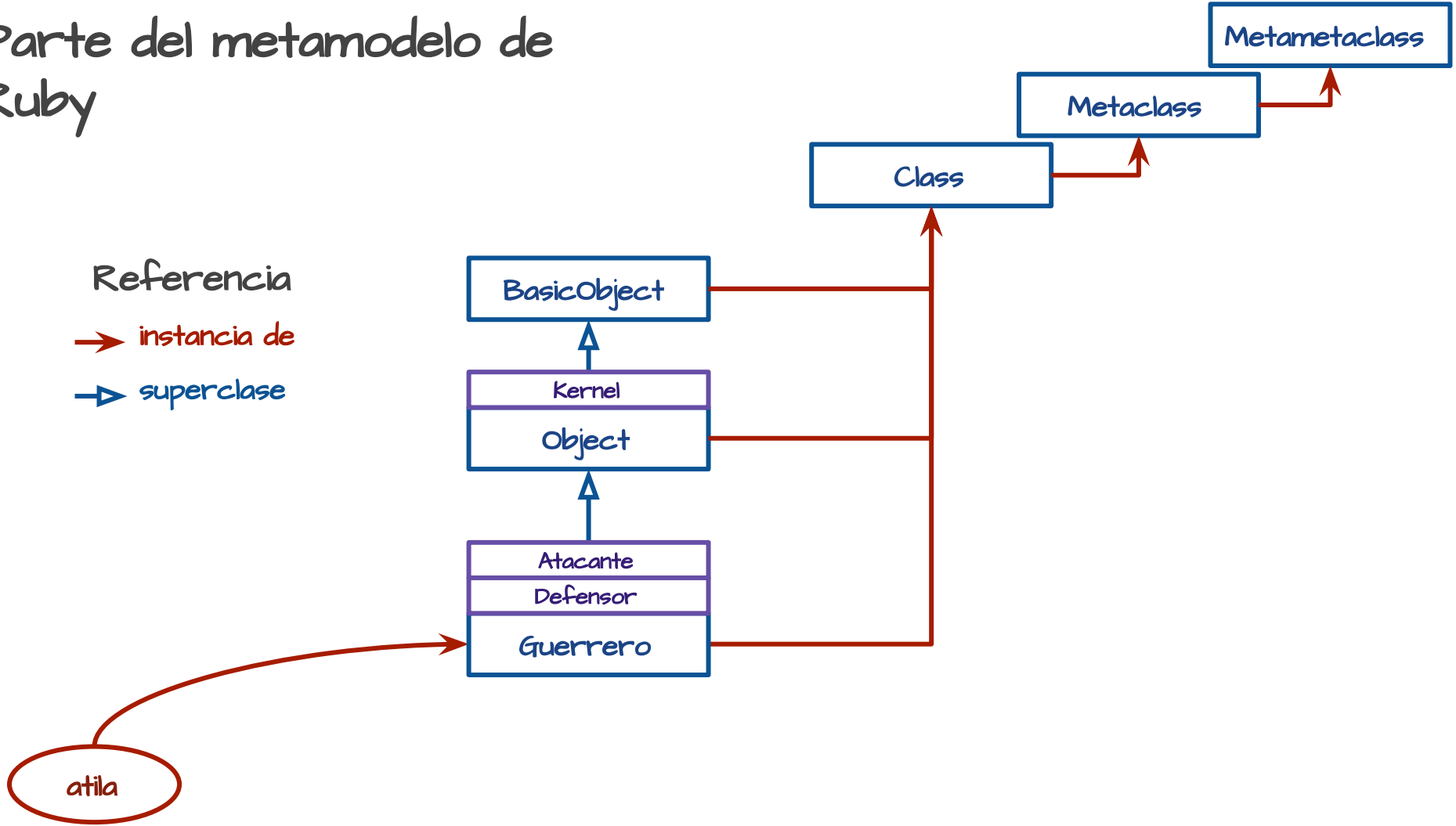

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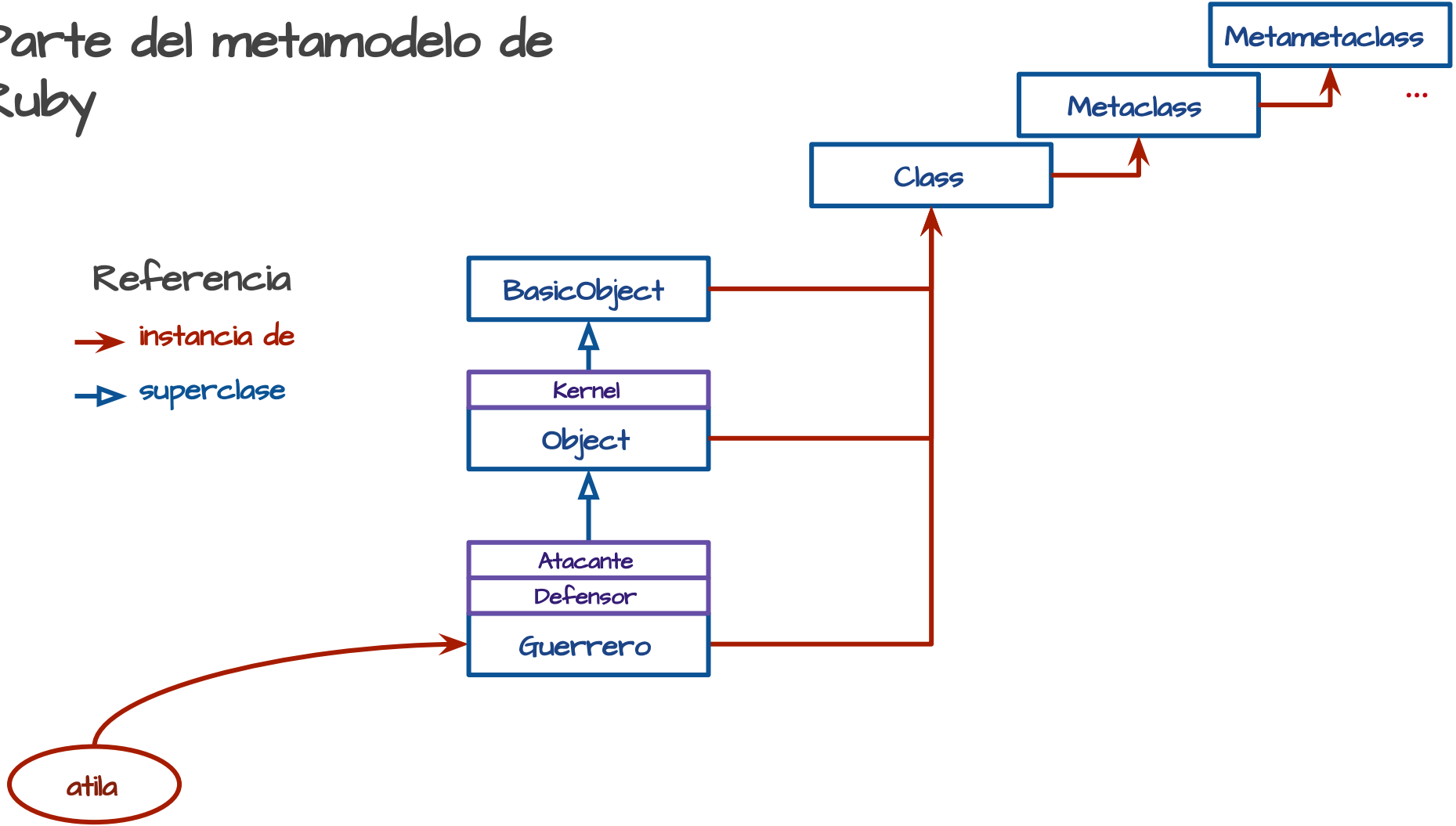

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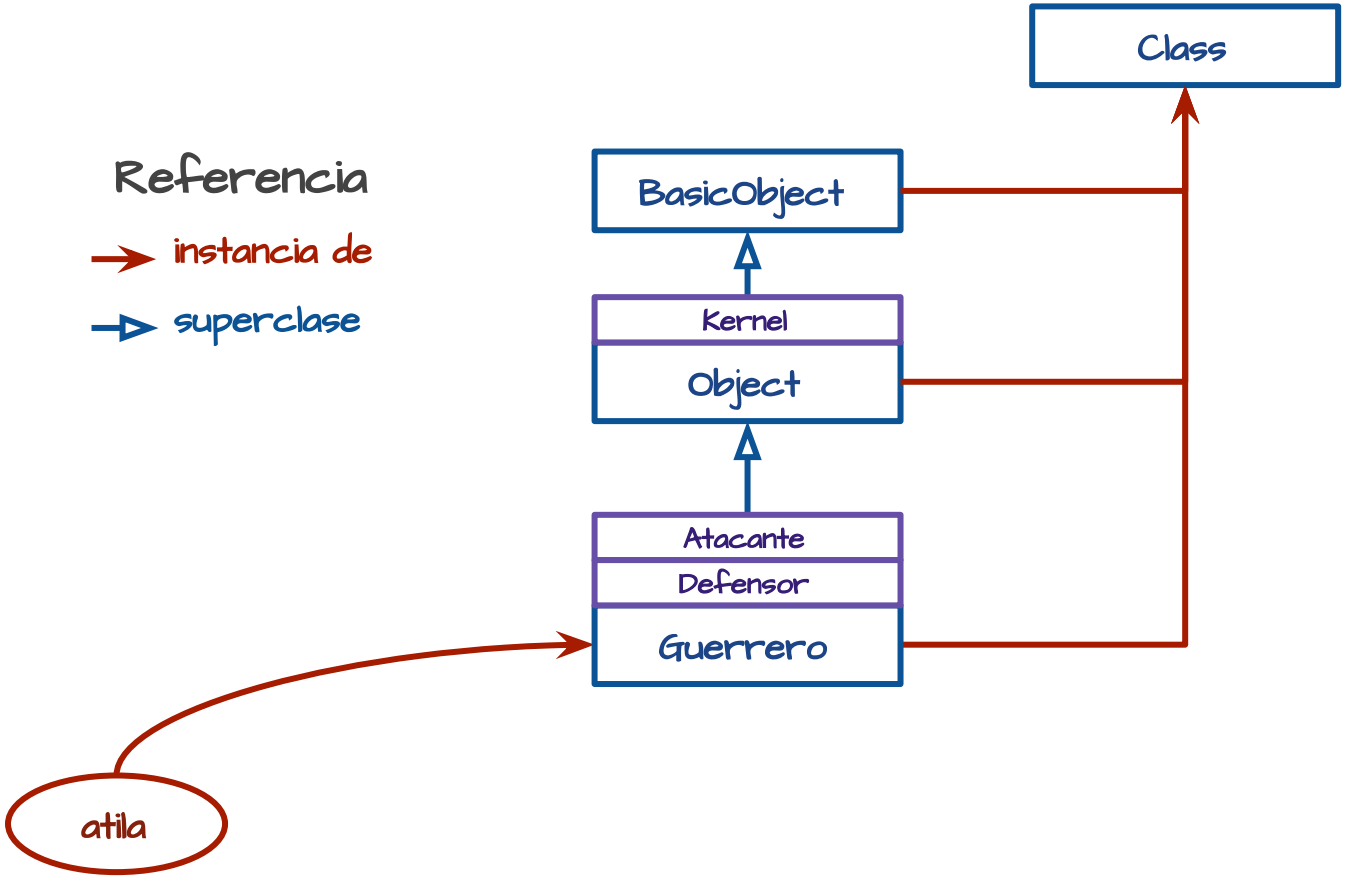
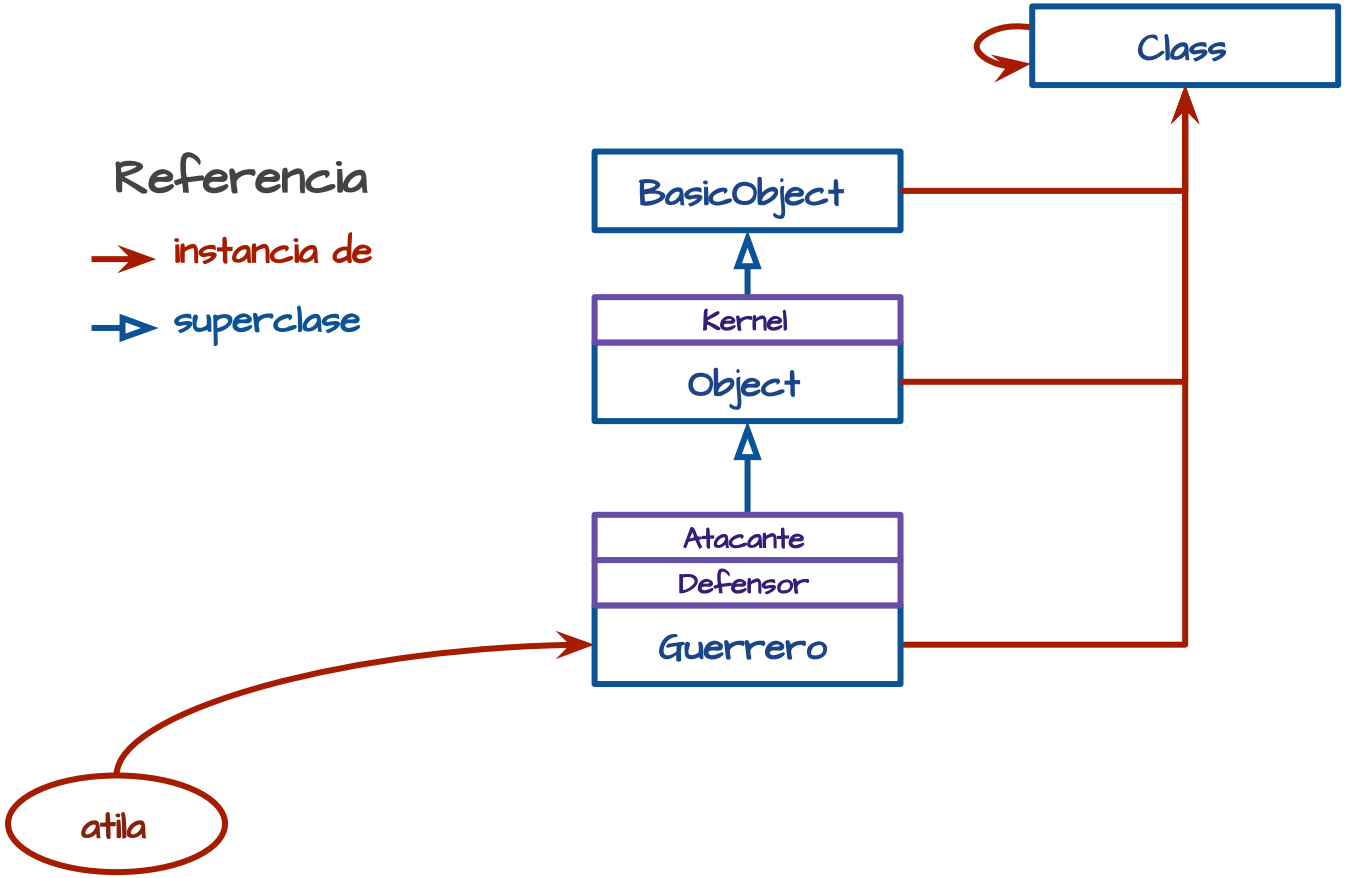


Parte del metamodelo de Ruby



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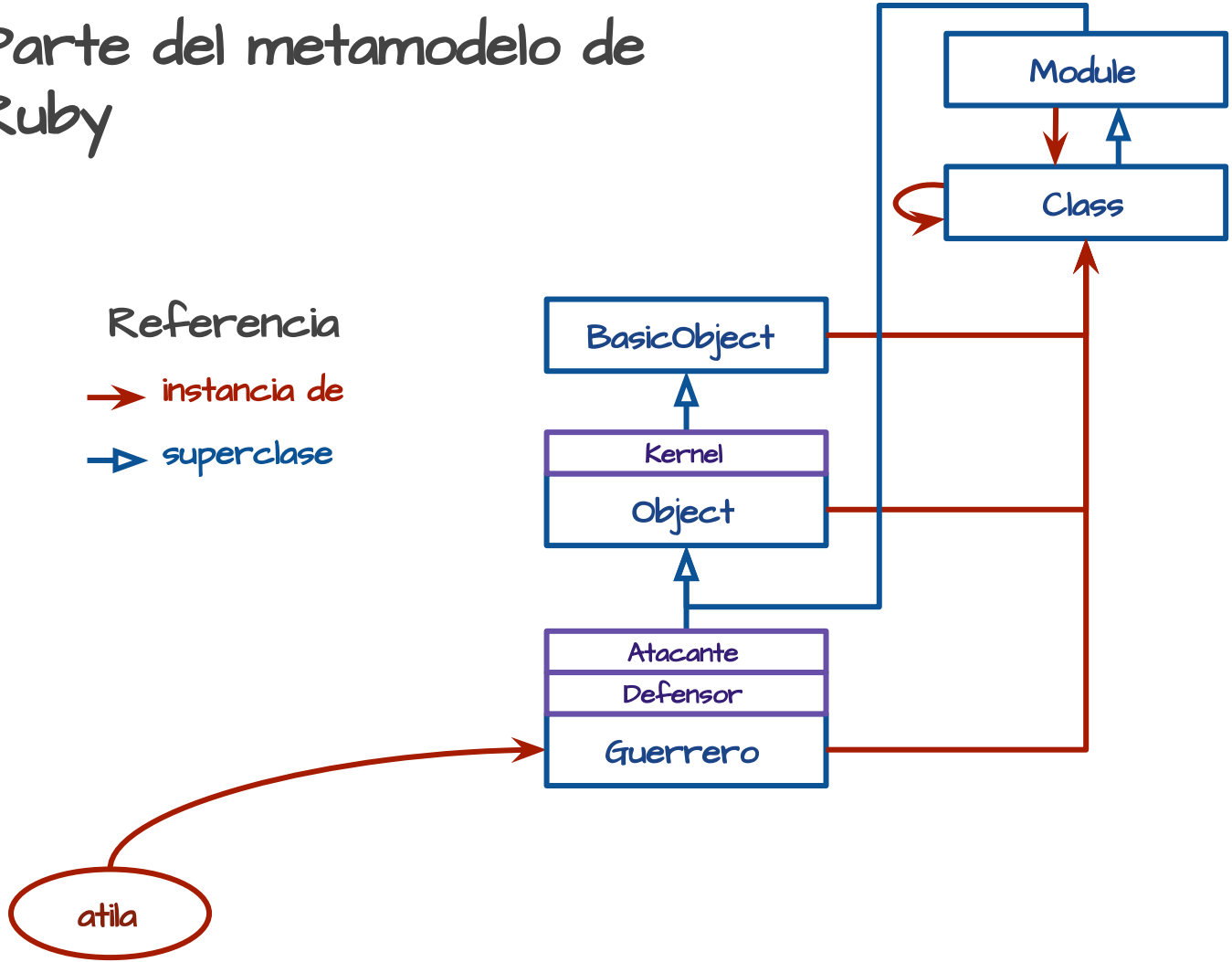
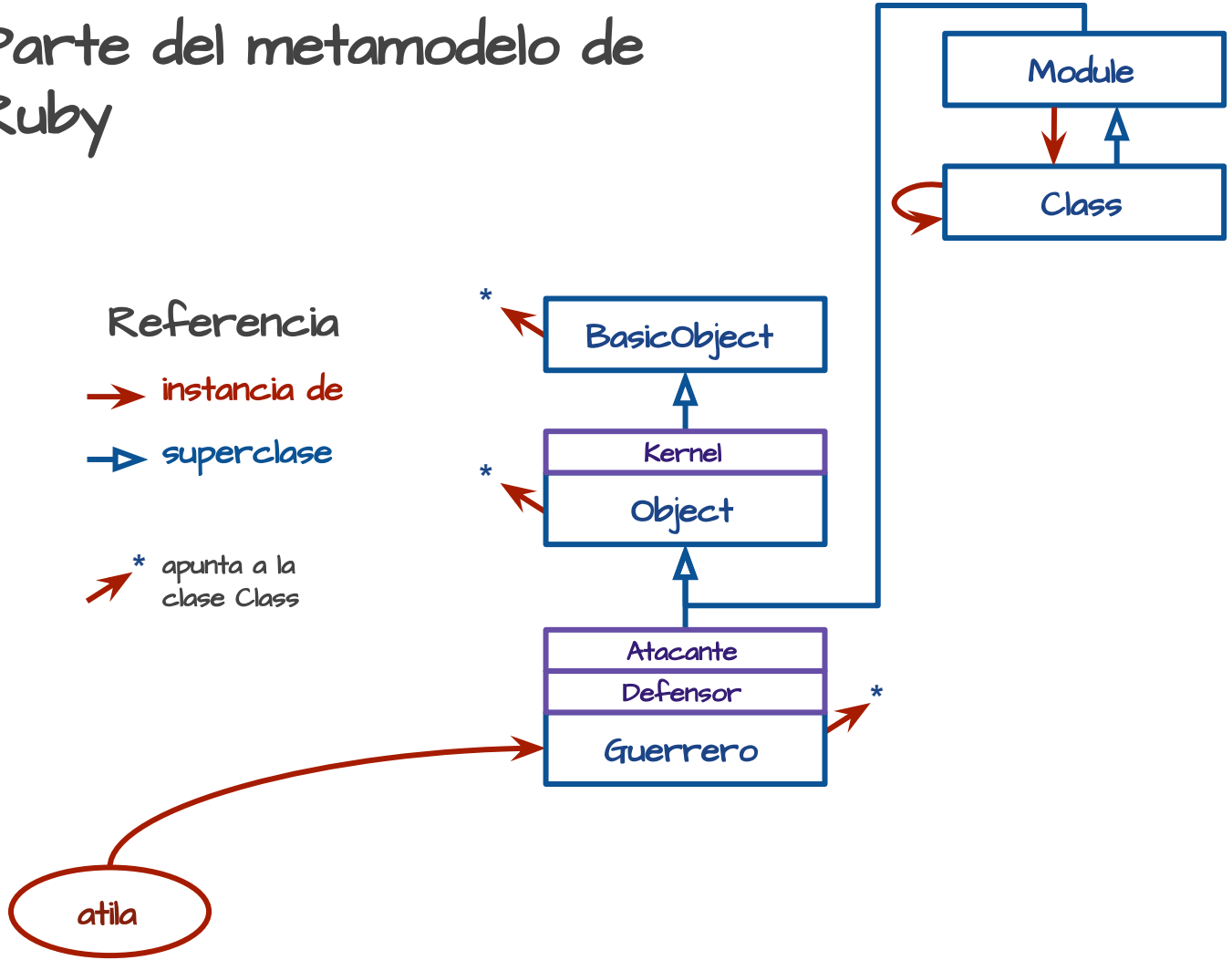
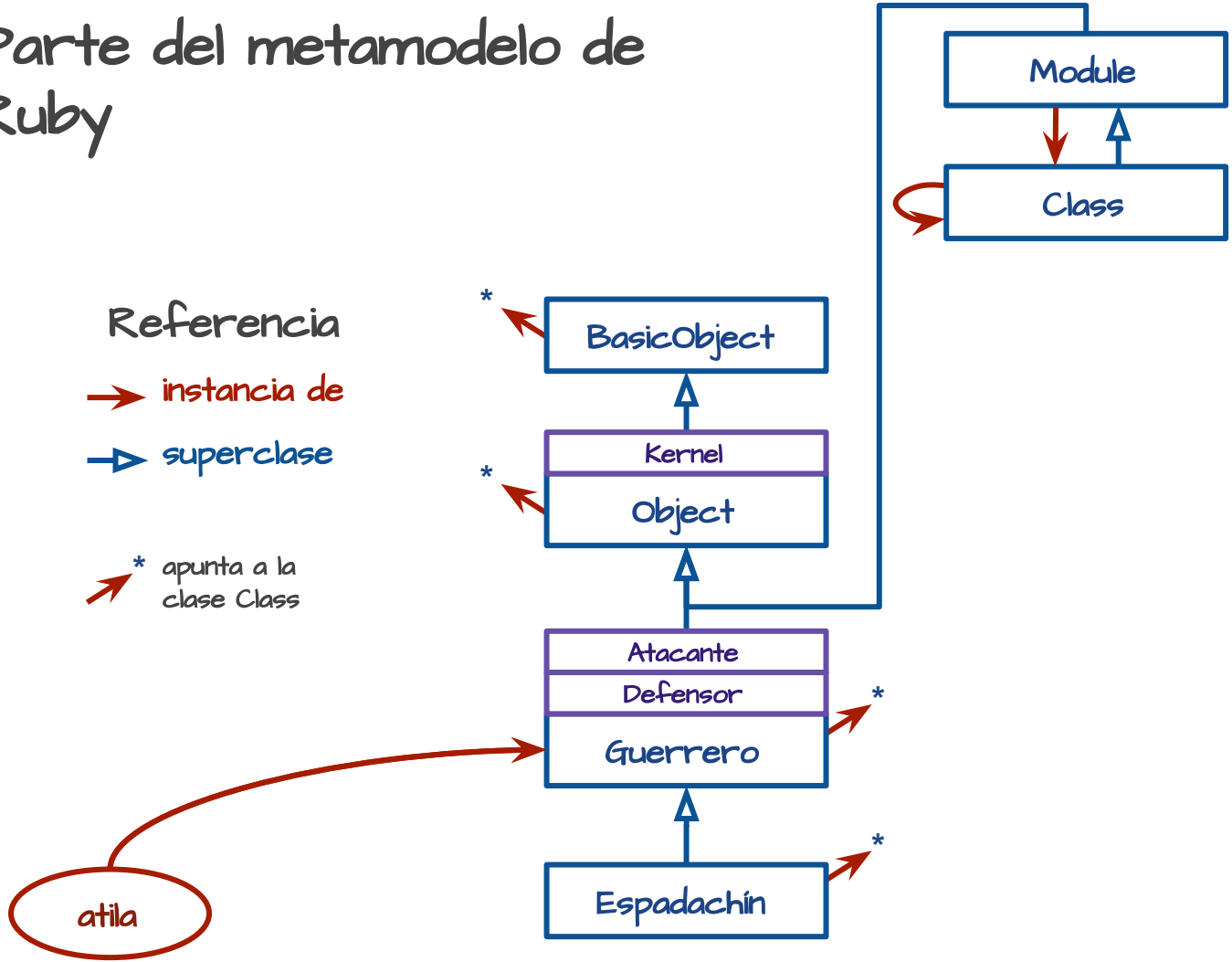


Parte del metamodelo de Ruby

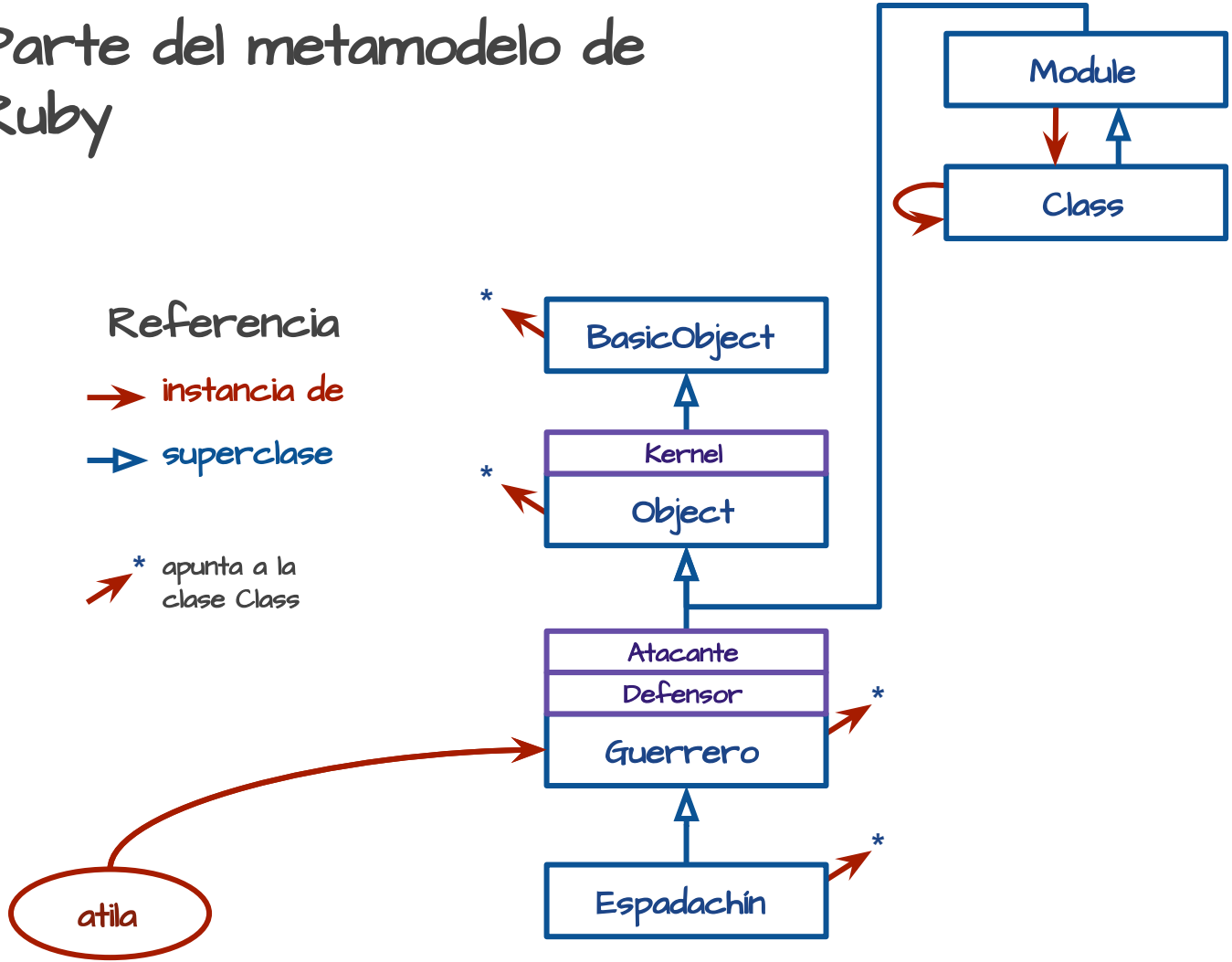


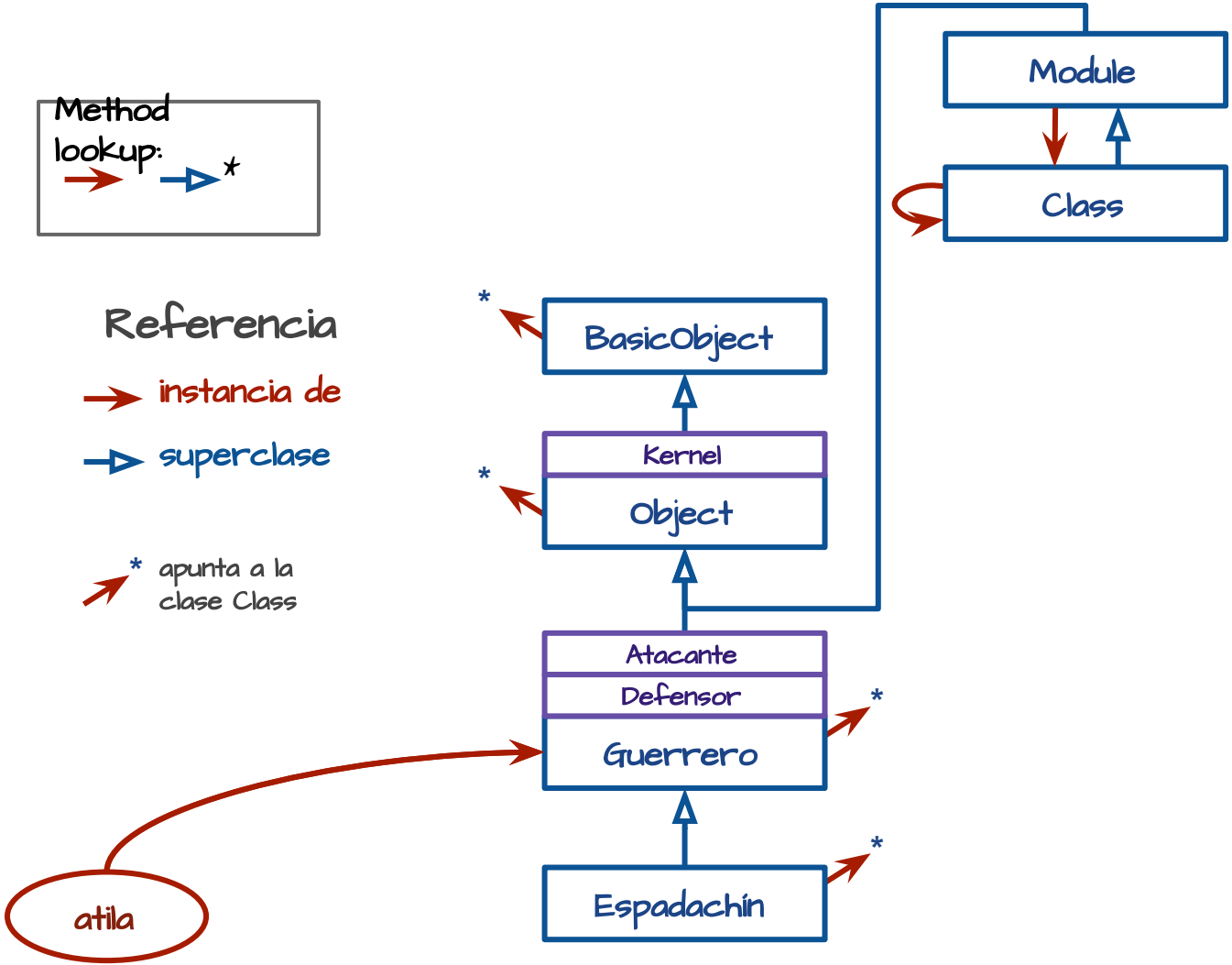
Parte del metamodelo de Ruby



Repaso: el proceso de búsqueda de métodos

Parte del metamodelo de Ruby




Method lookup:
*
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instancia de
superclase
* apunta a la clase Class
Module
Class
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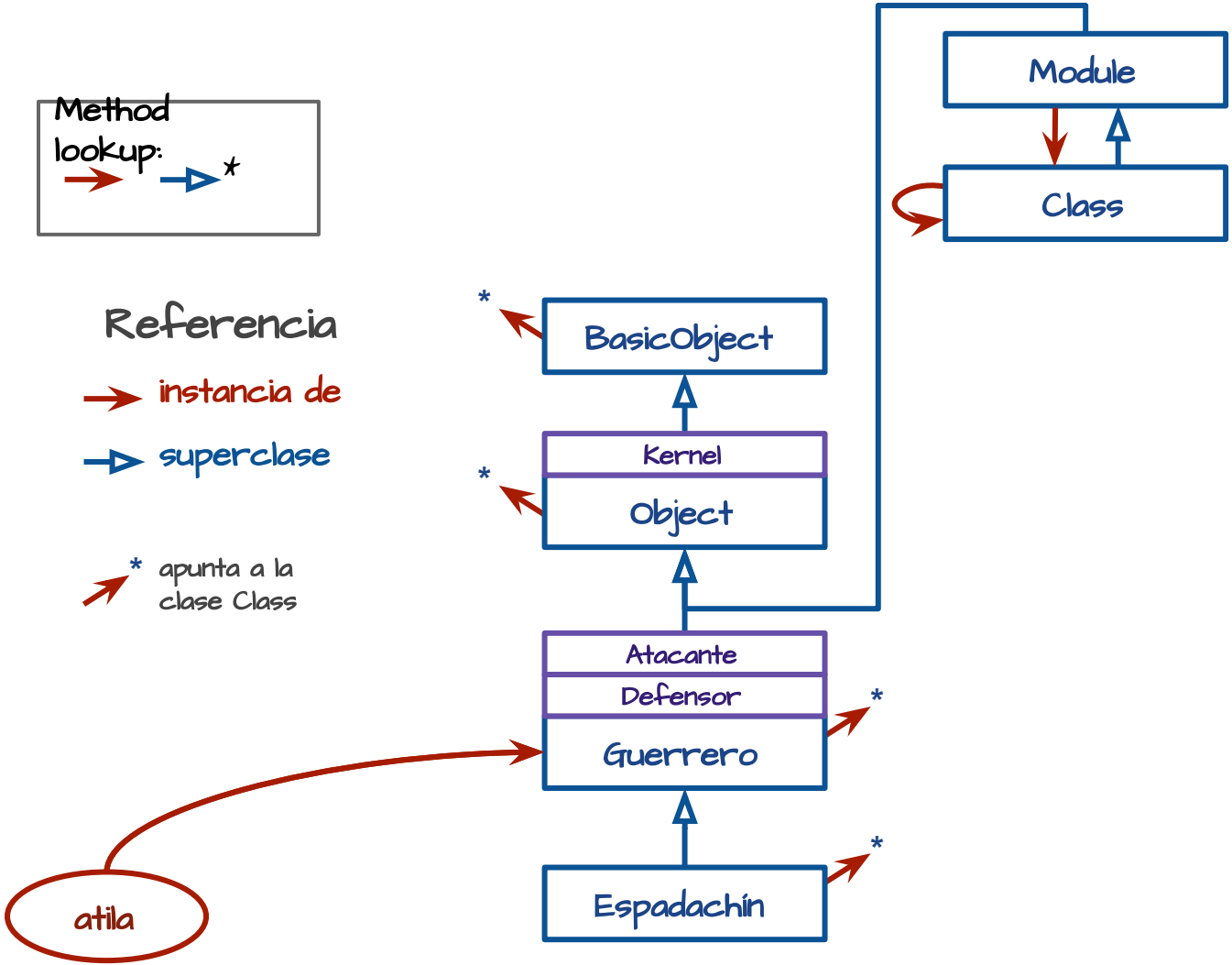
atila

¿Cómo hacen las clases para entender mensajes específicos (“mensajes de clase”) en Ruby?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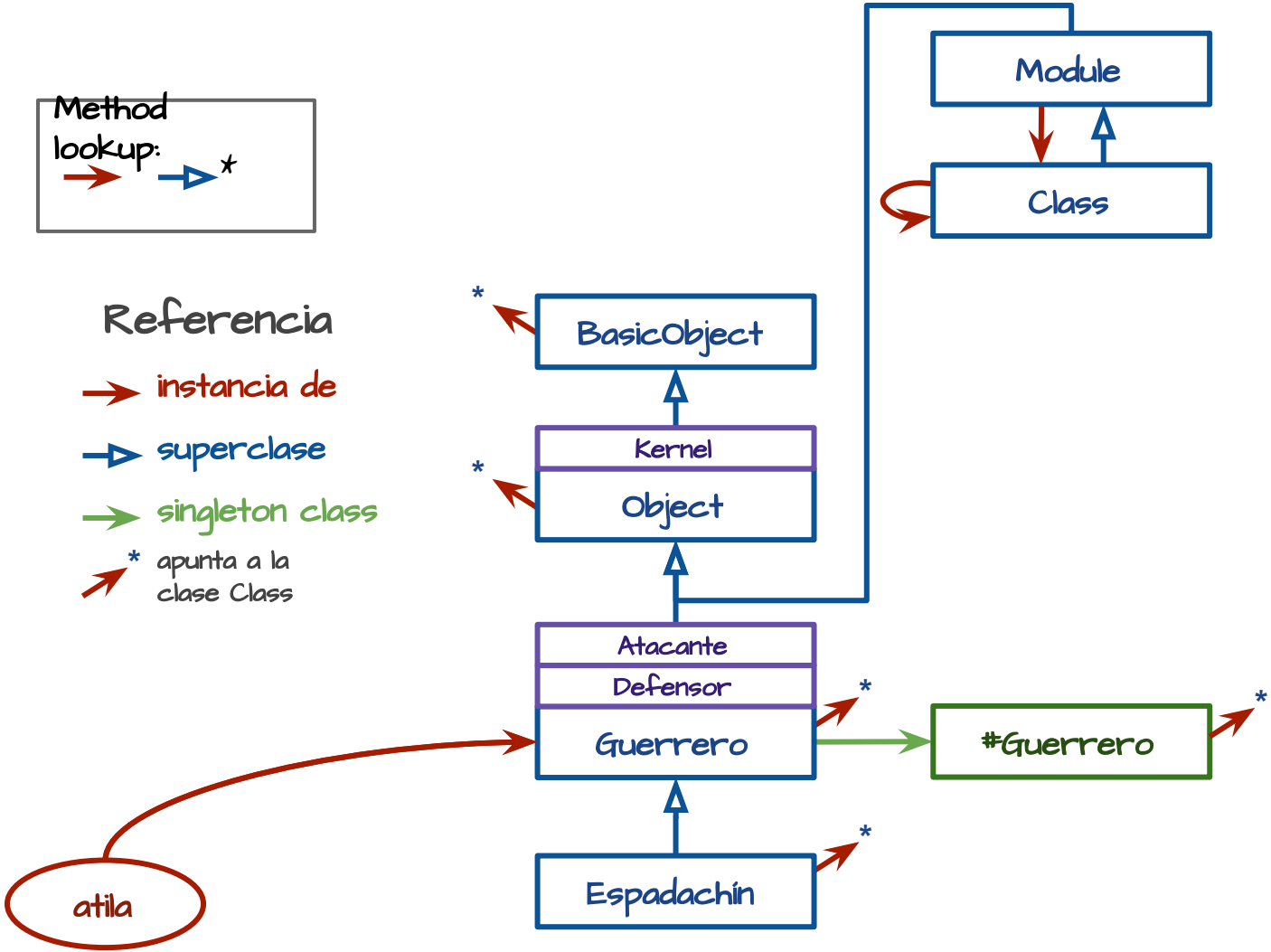
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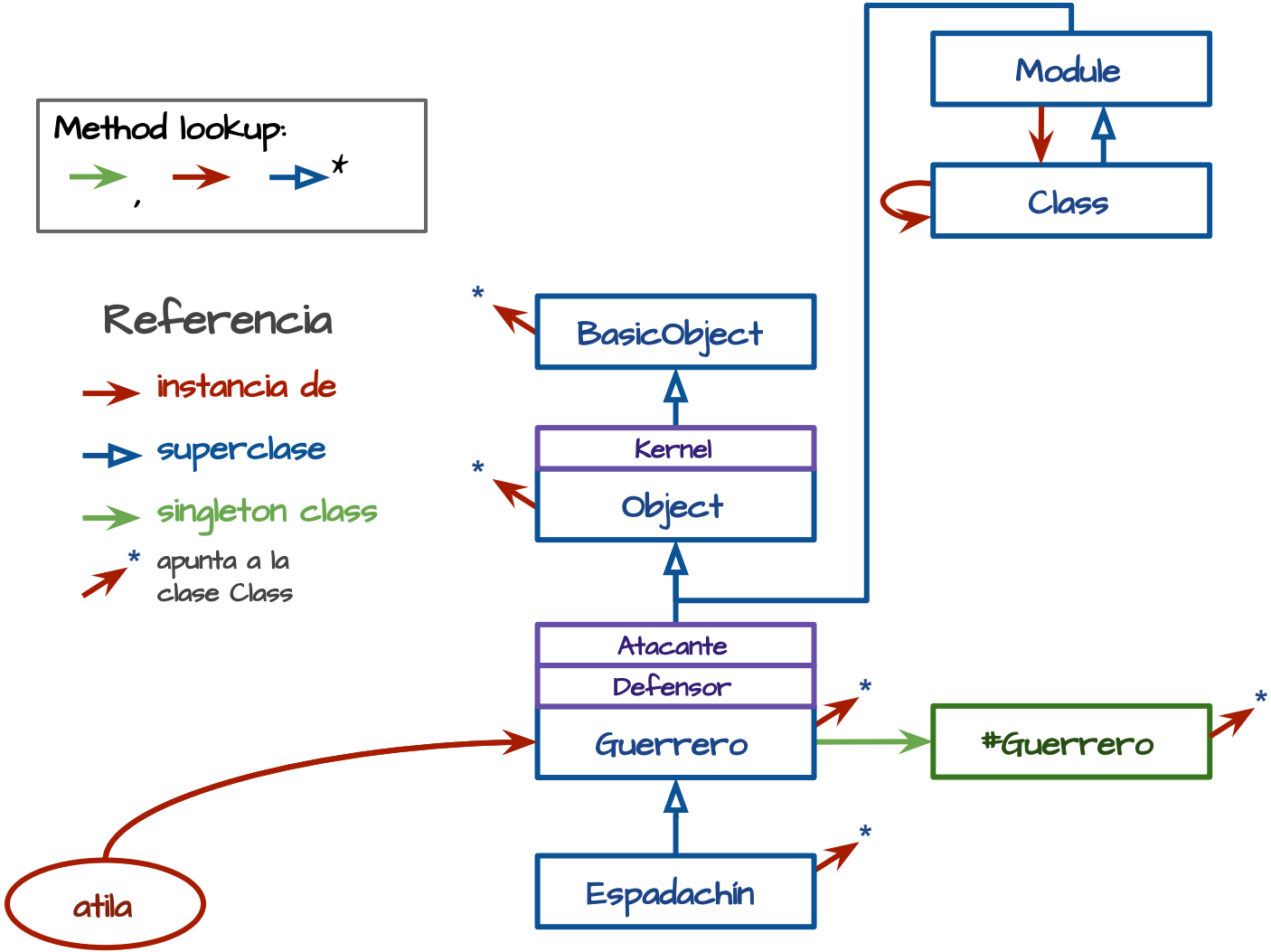
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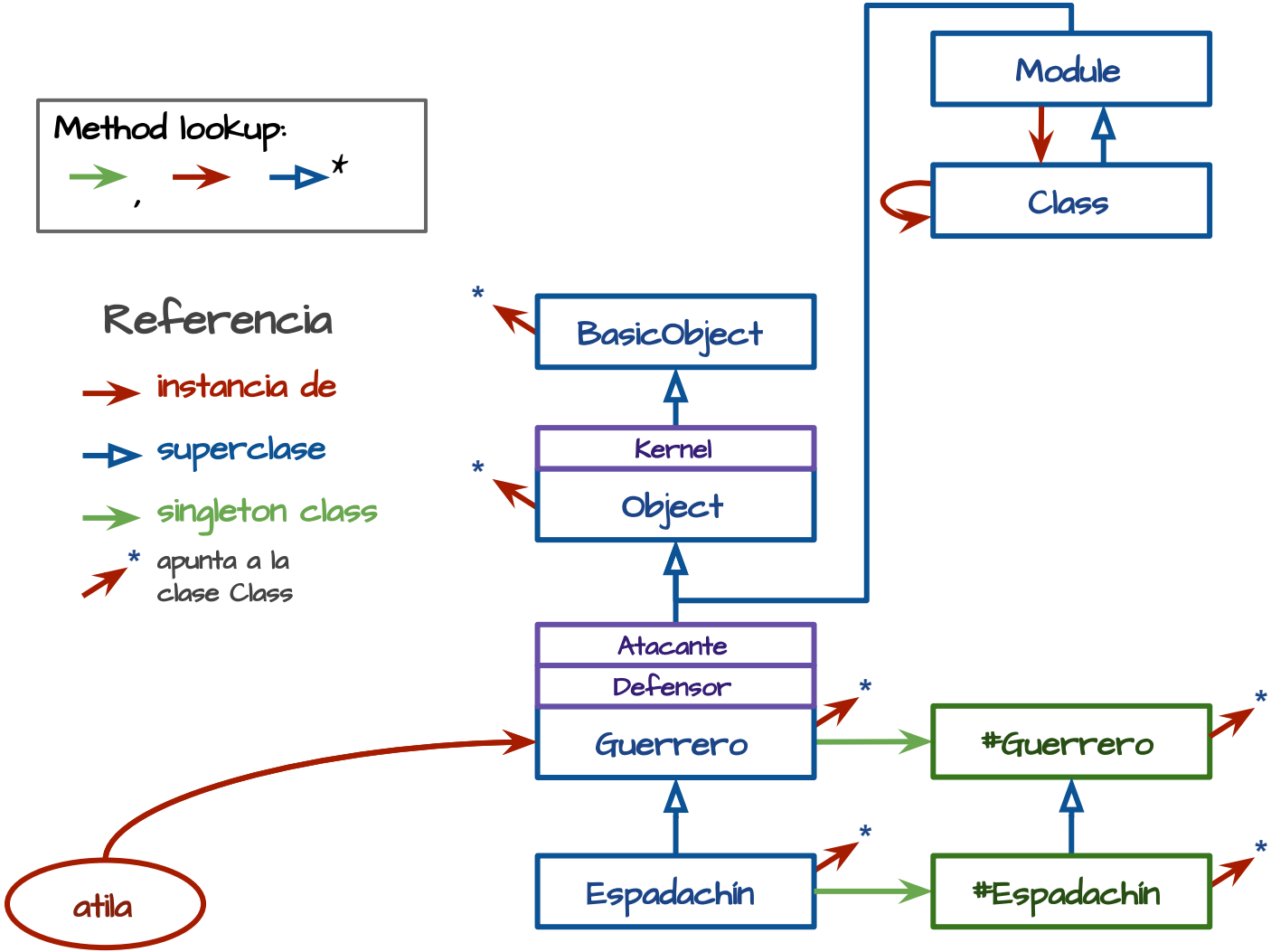
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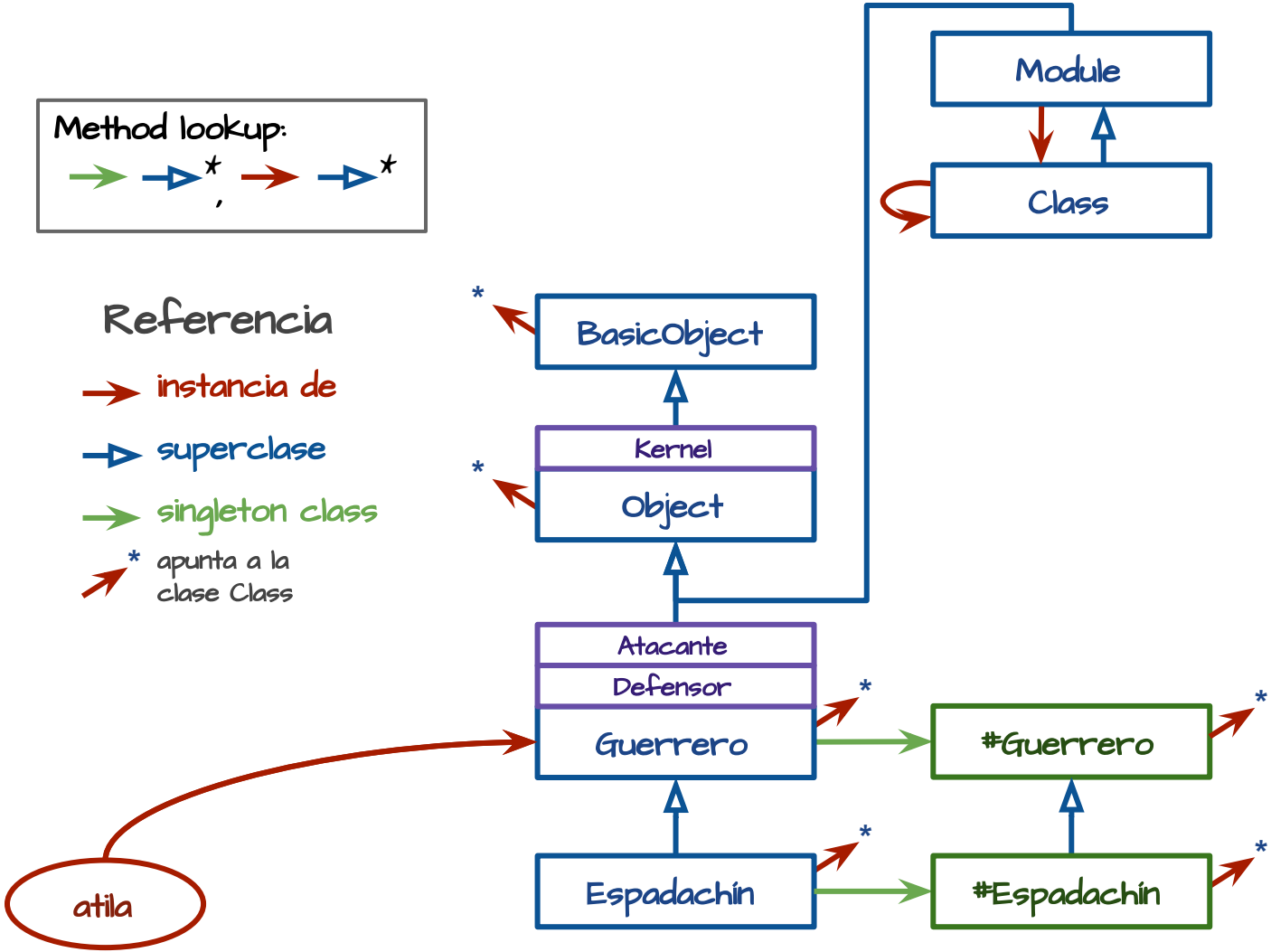
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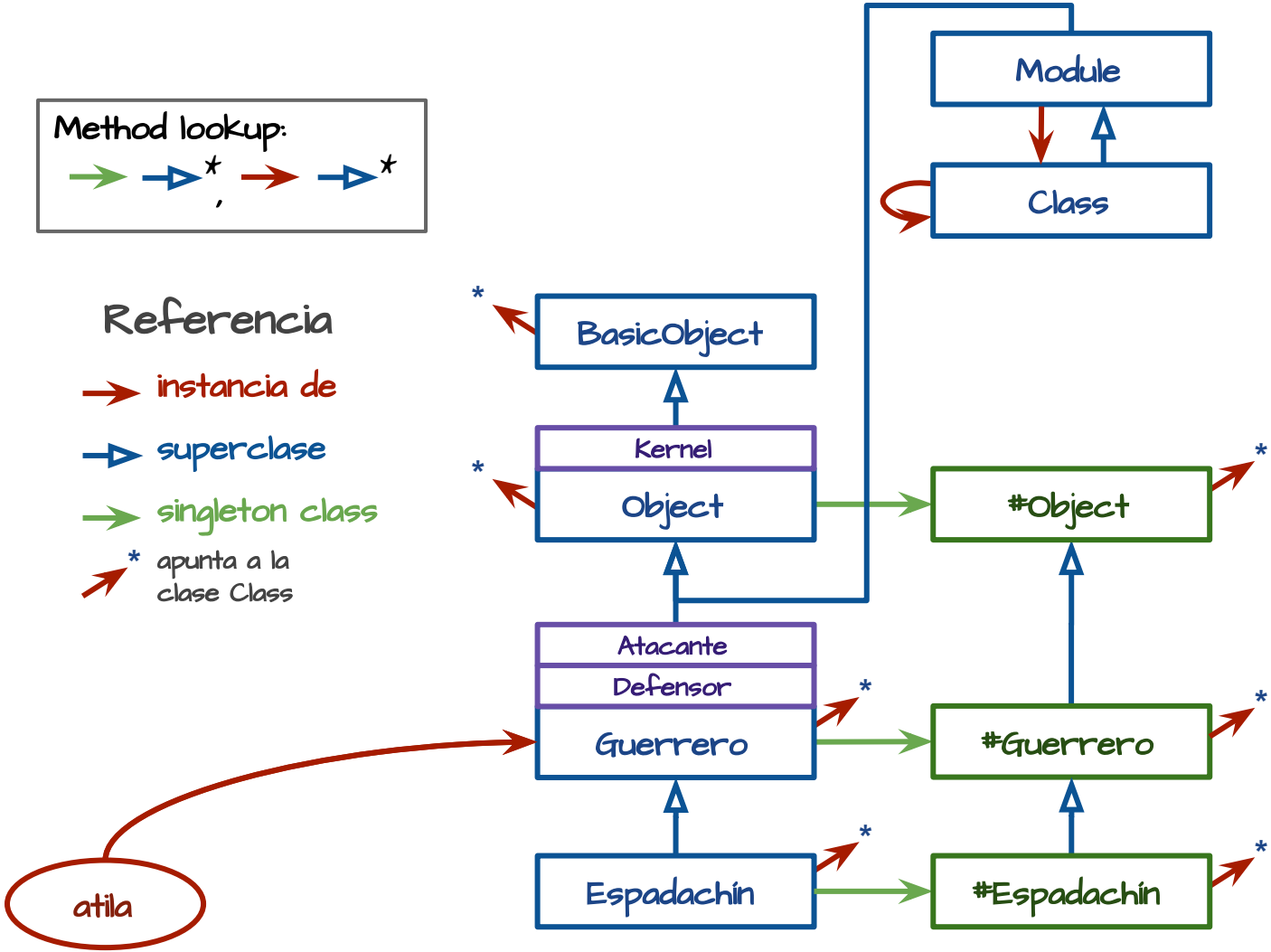
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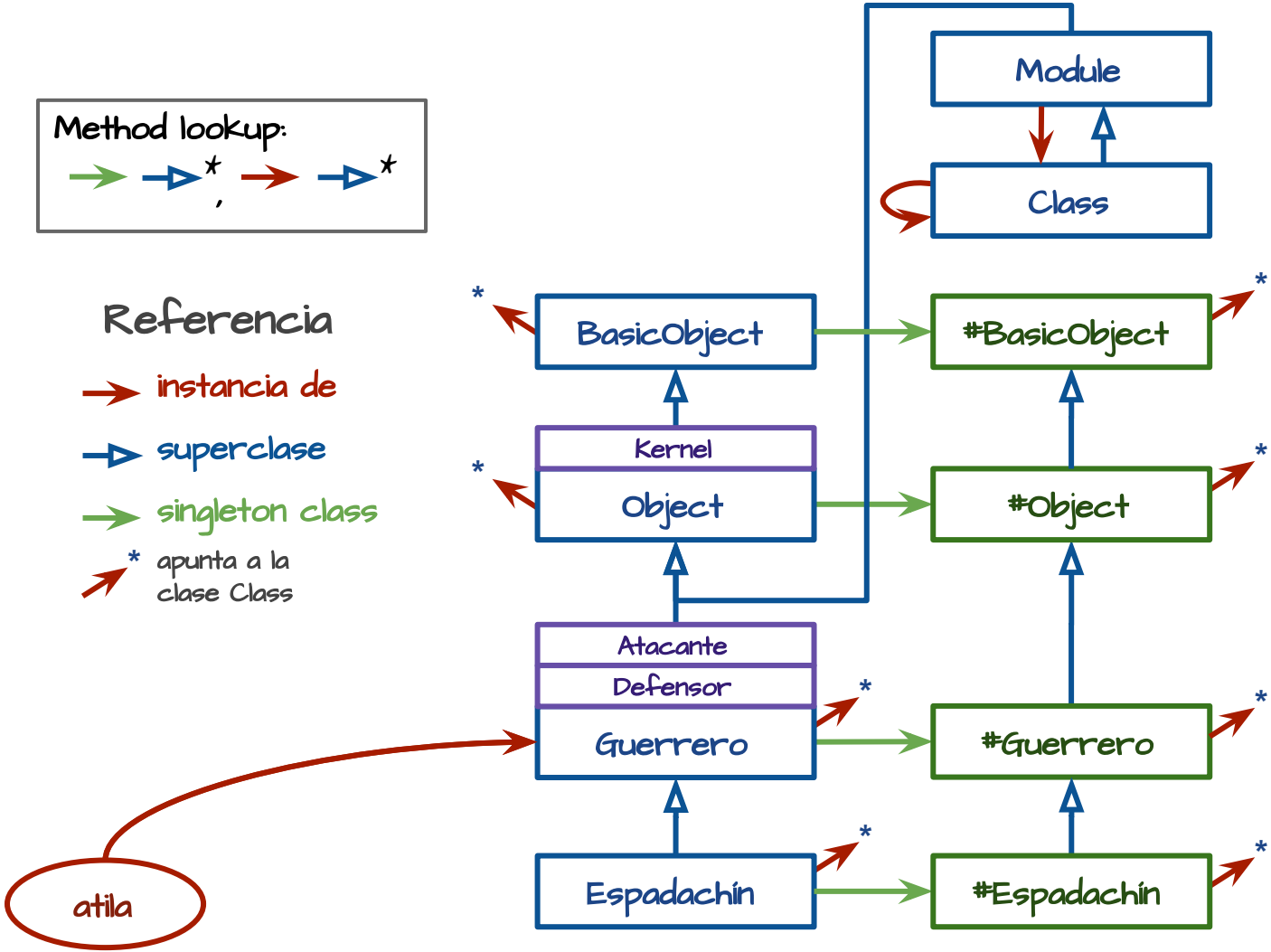
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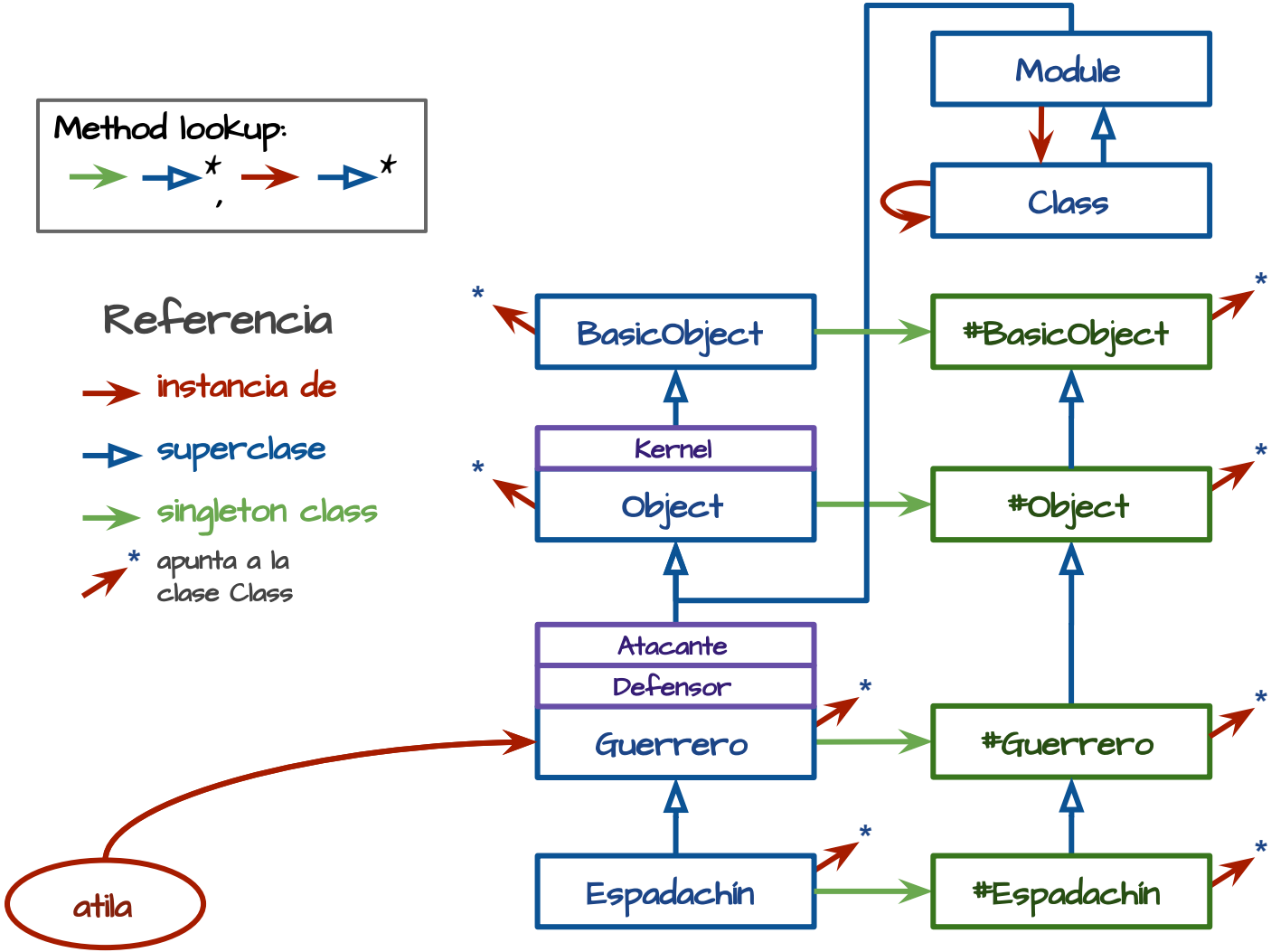
¿Podemos simplificar el *method lookup*?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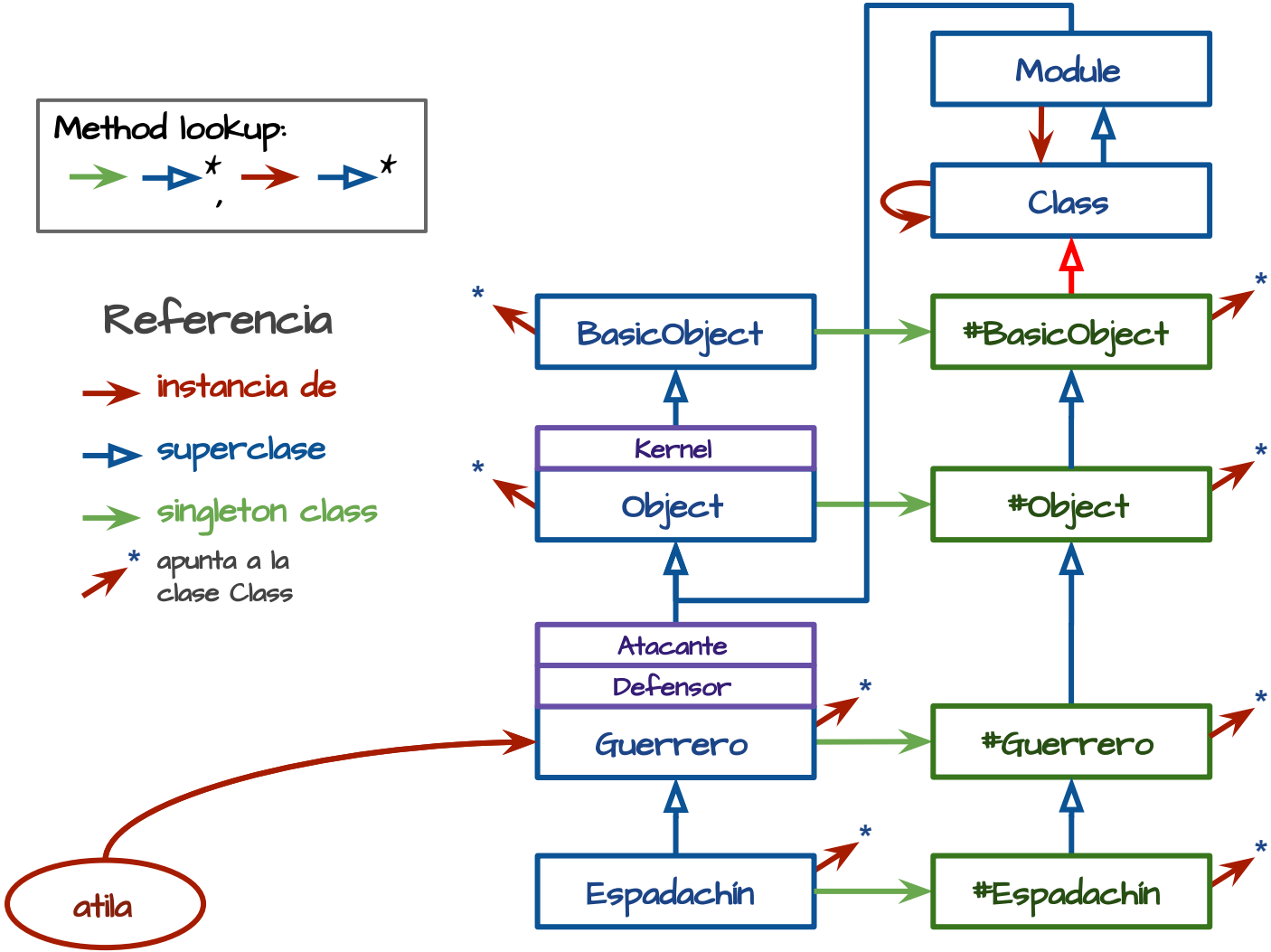
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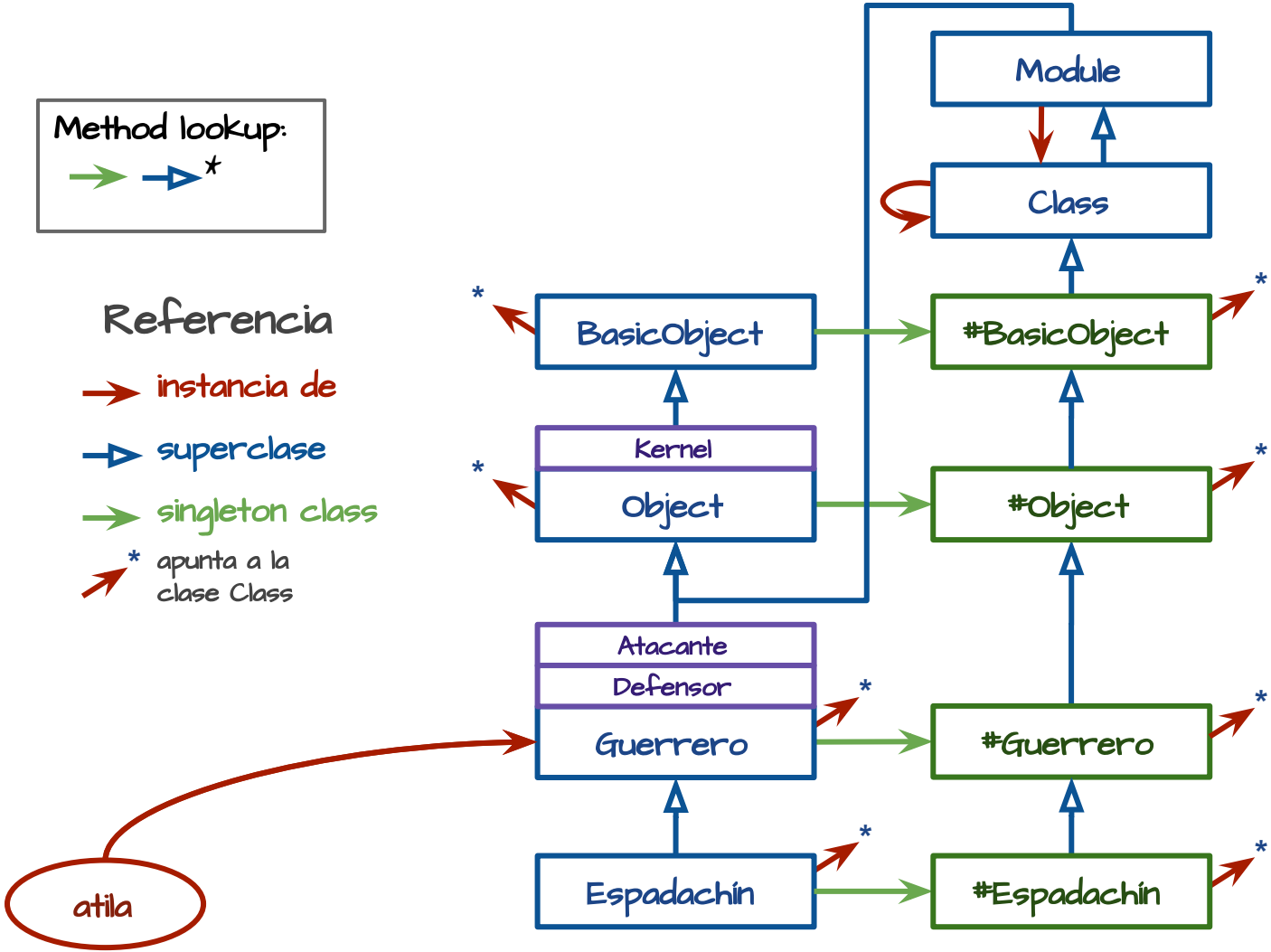
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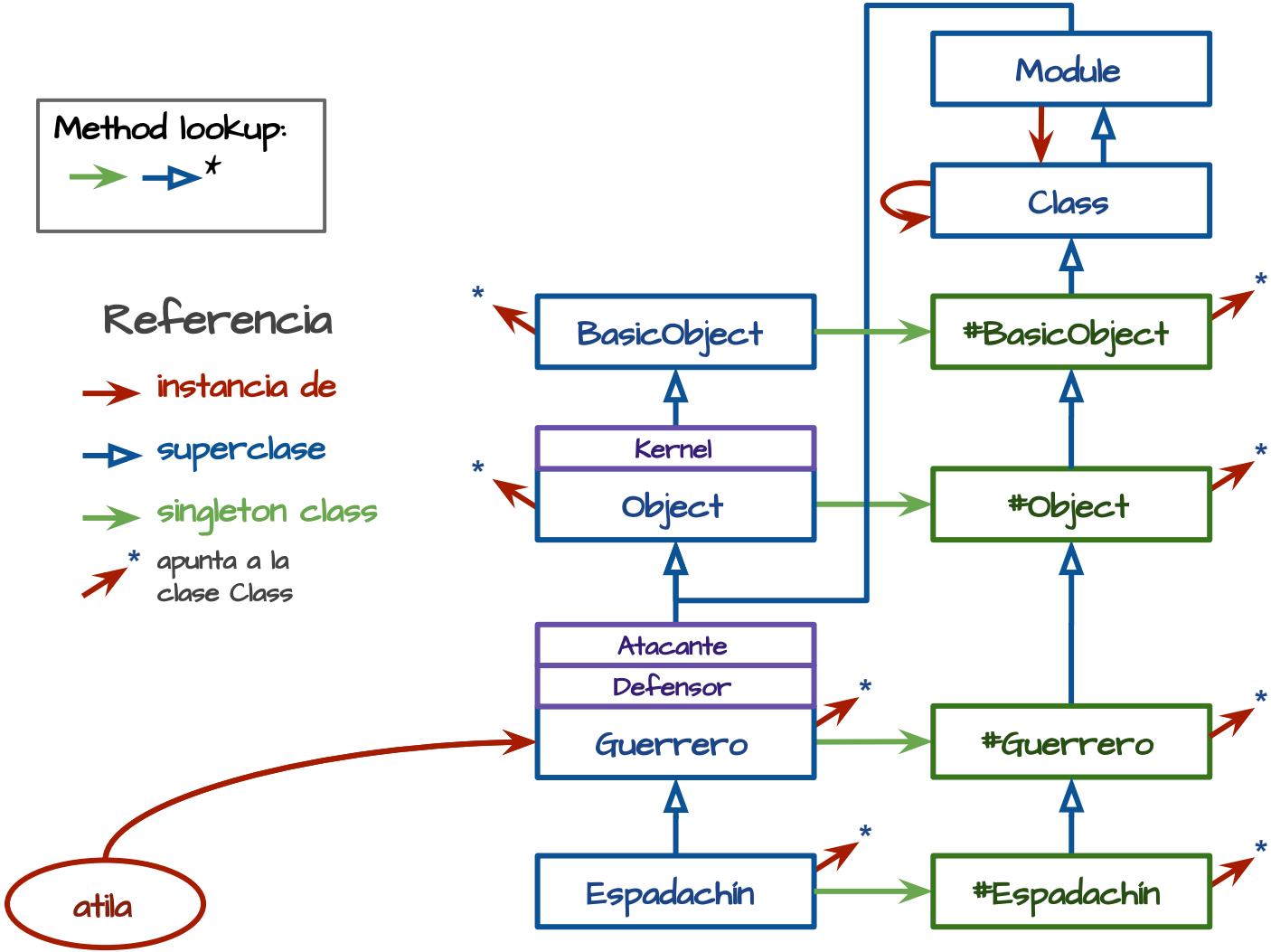
El camino a partir de la *singleton class* **agrega un prefijo** (estricto) al camino a partir de la respuesta a *class*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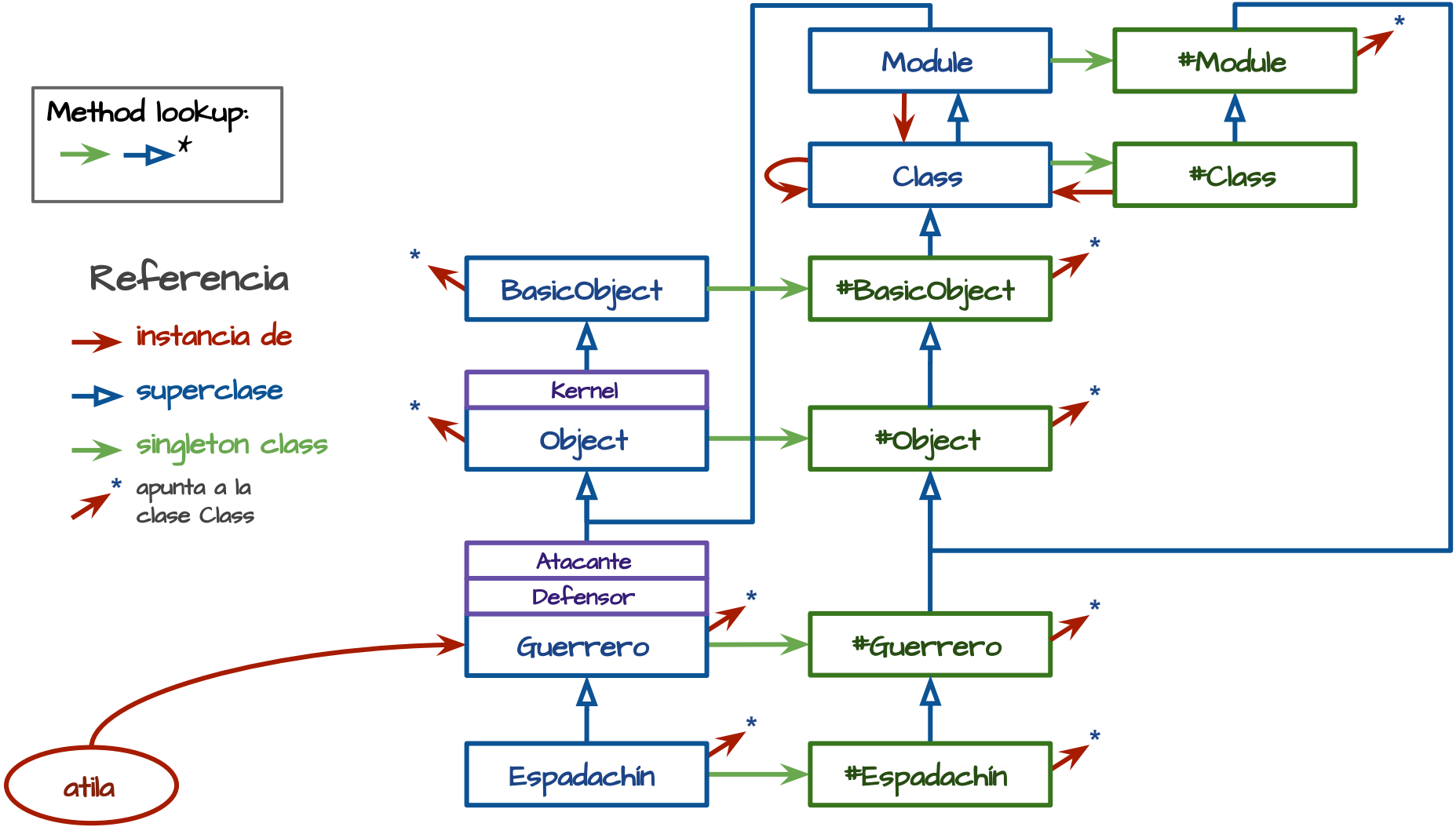
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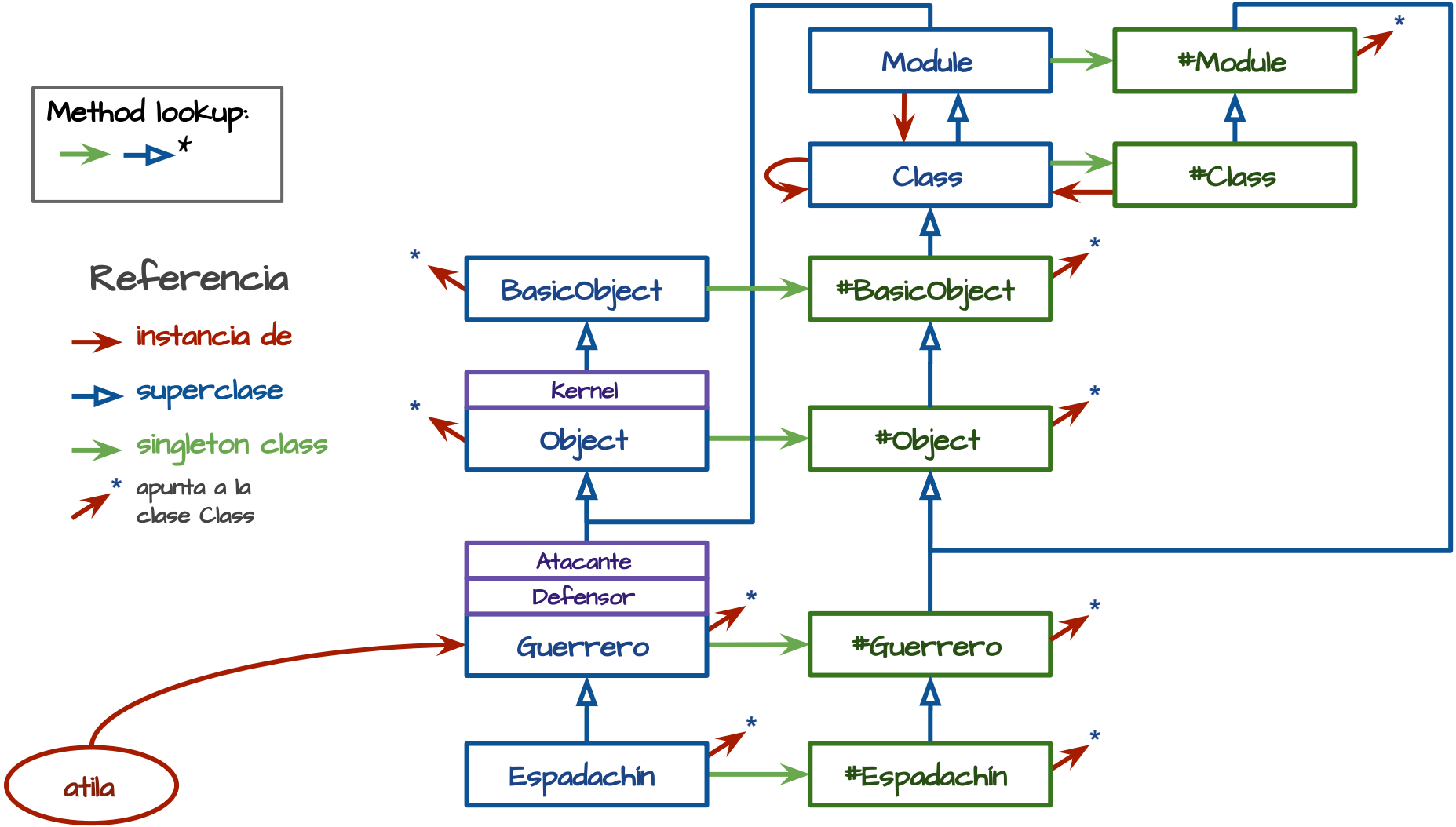
Pero... una *singleton class* también es una clase!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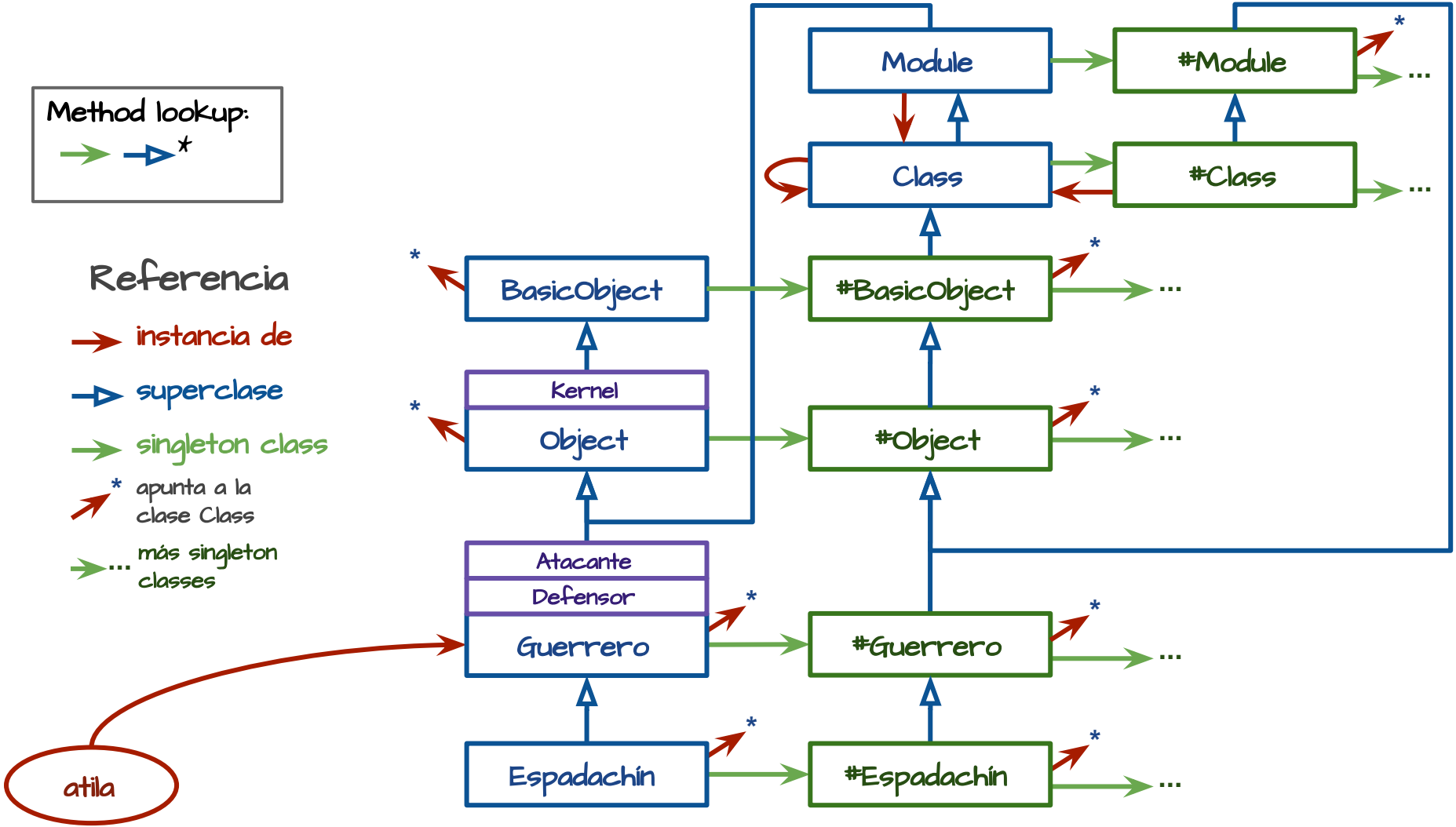
Method lookup:
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... más singleton classes
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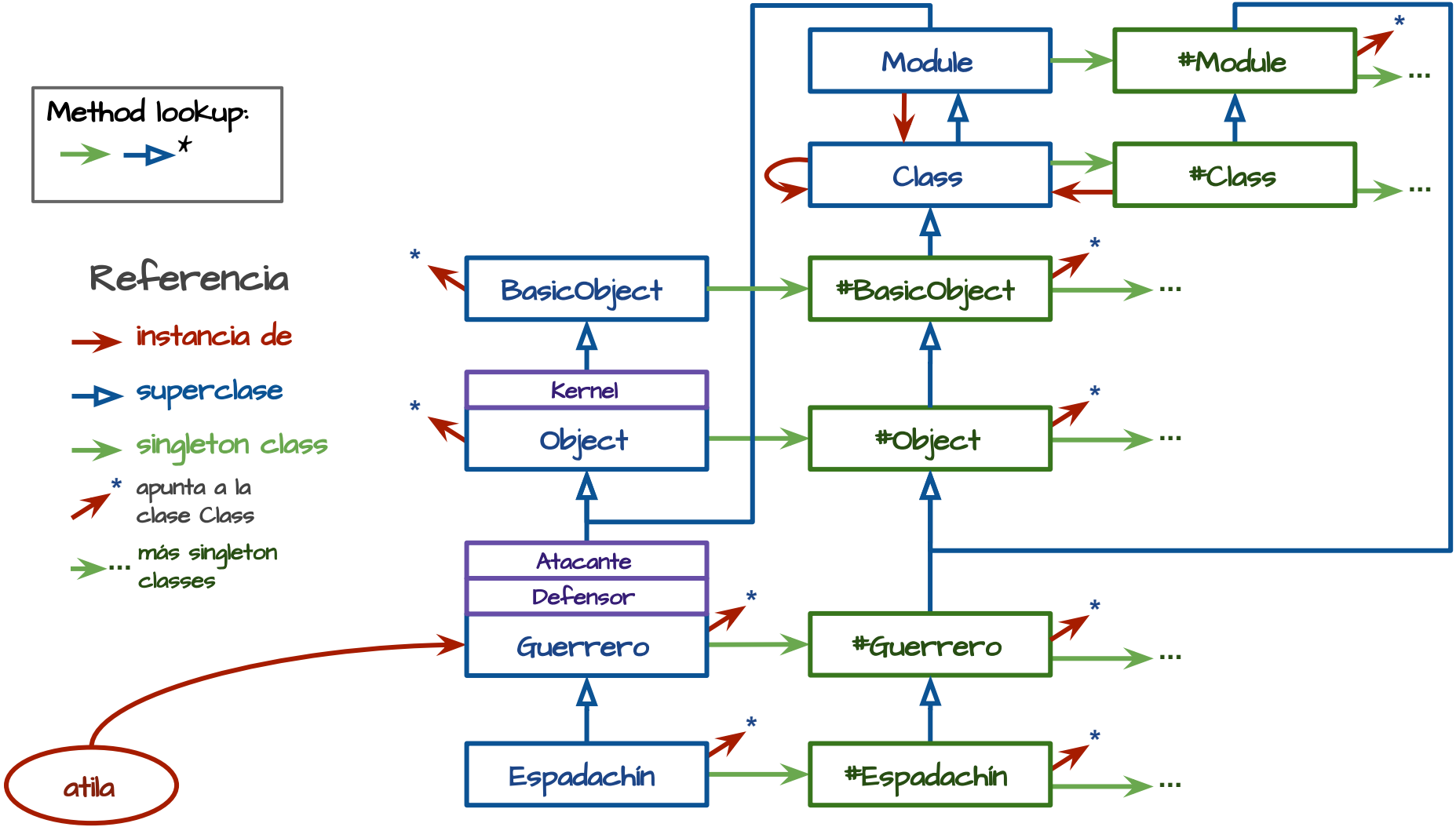
¿Qué pasa con el *method lookup* cuando un objeto (que no es una clase) recibe un mensaje?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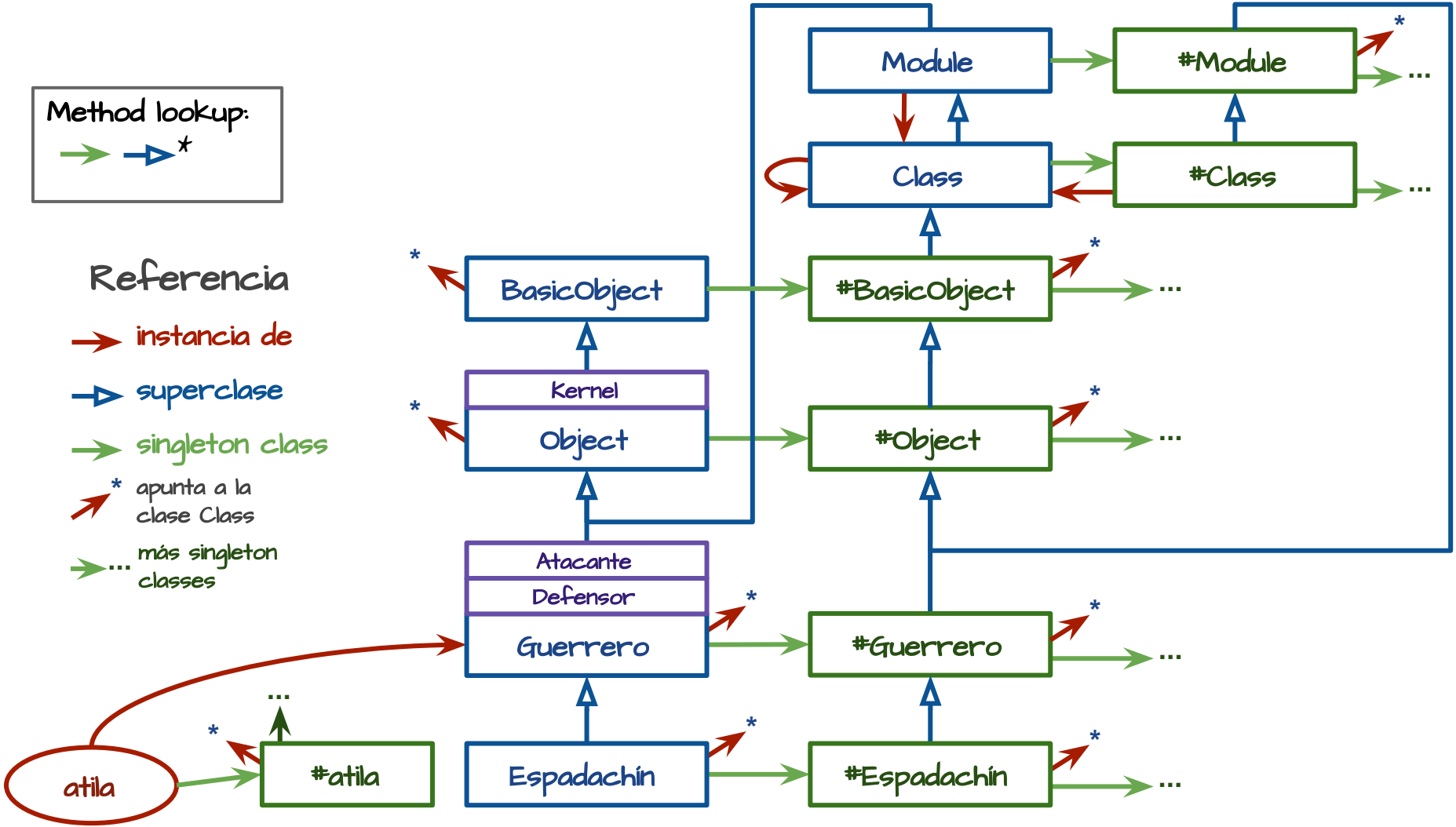
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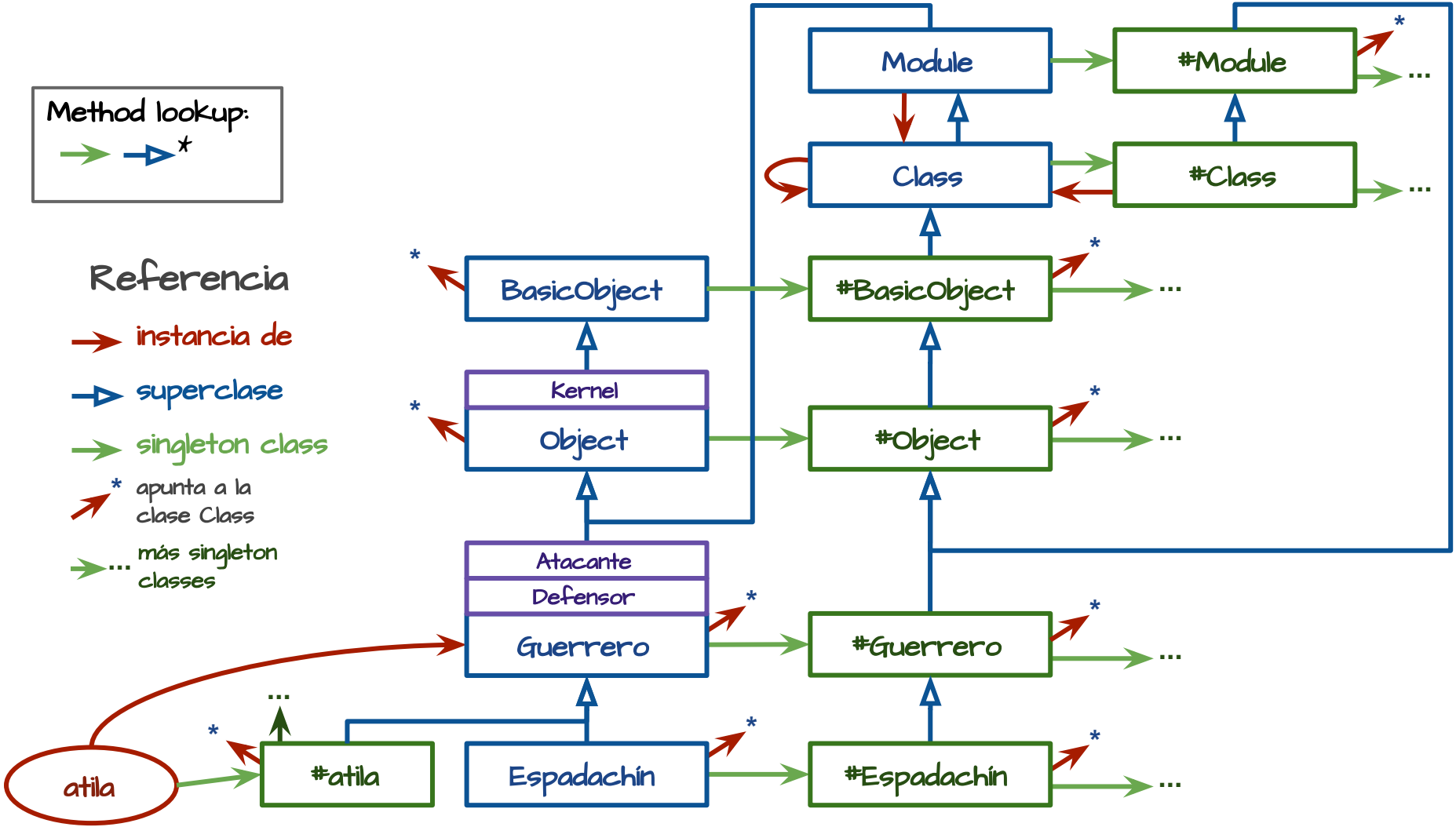
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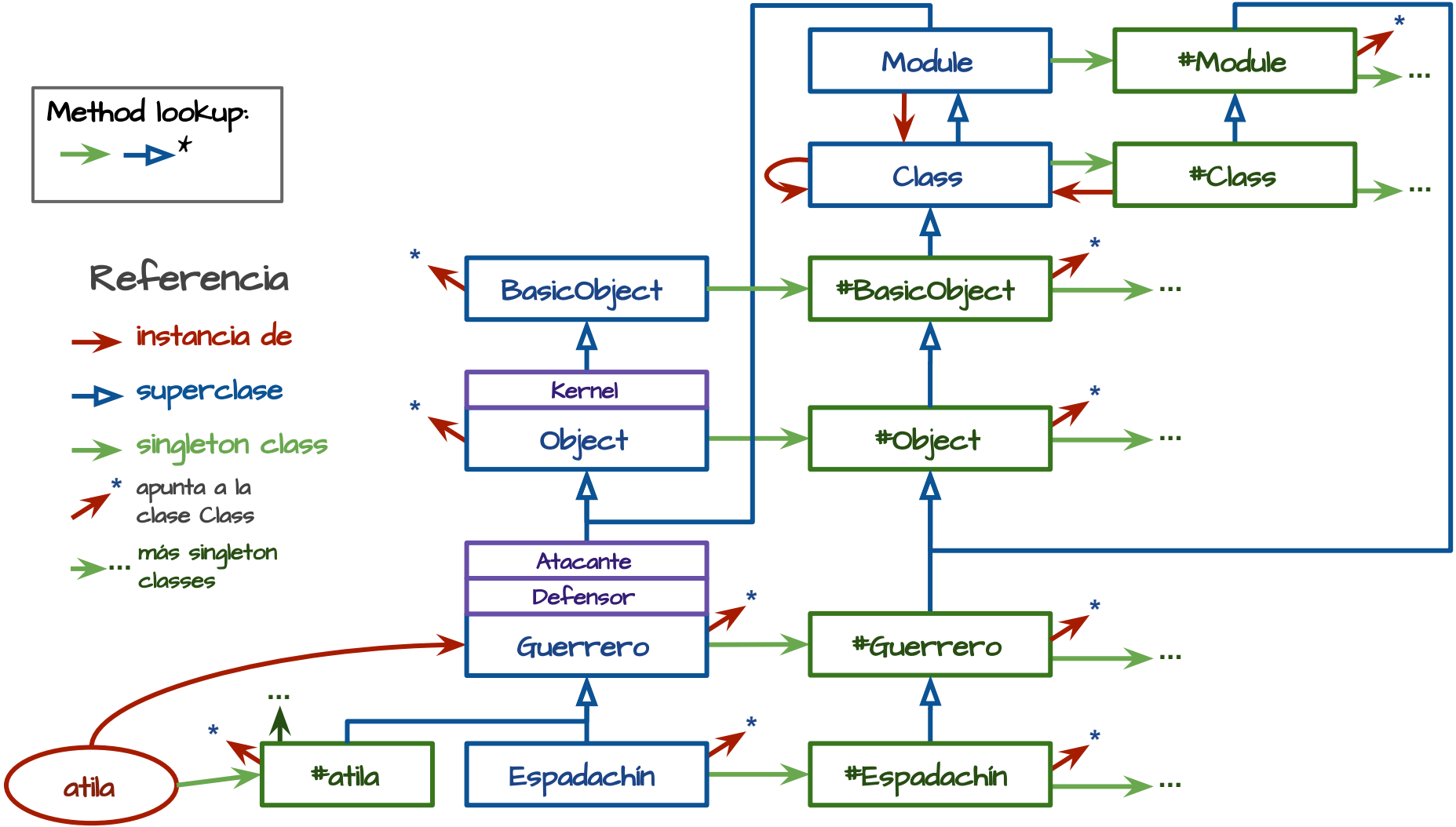
Se sigue cumpliendo que el camino a partir de la *singleton class* **agrega un prefijo** al camino a partir de la respuesta a *class*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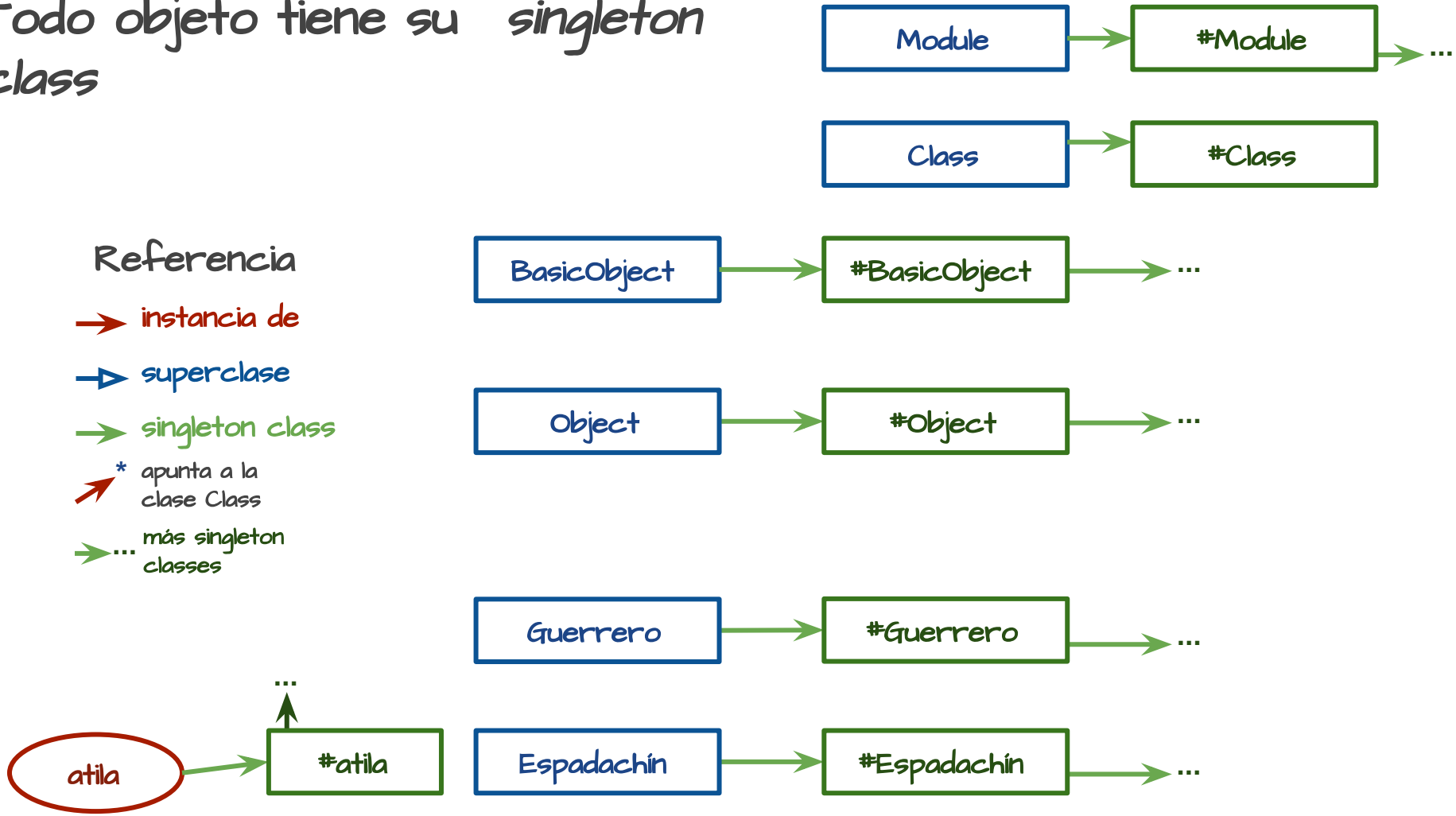
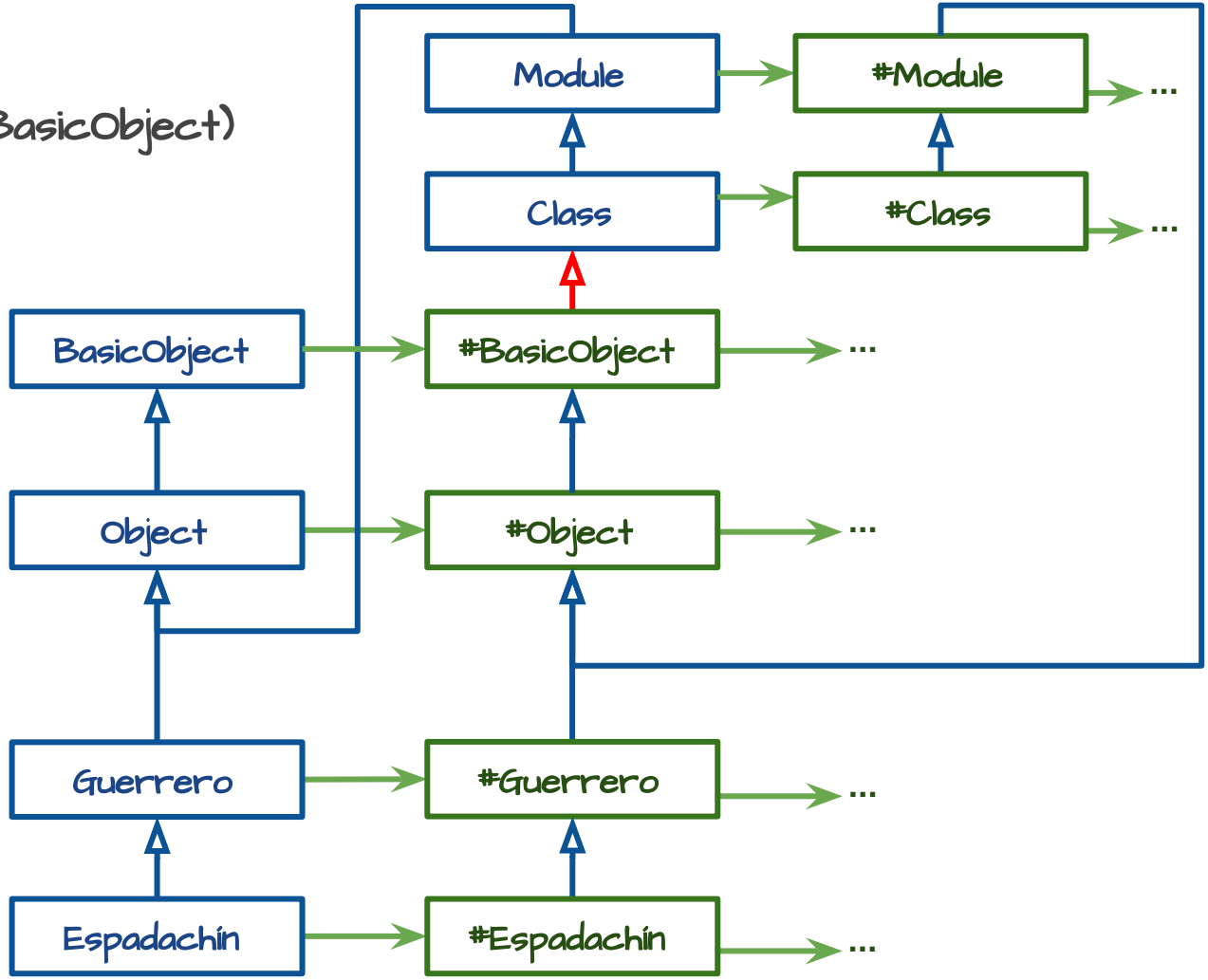
Algunas regularidades en el metamodelo

Todo objeto tiene su *singleton class*



Jerarquías paralelas

(excepto en el caso de BasicObject)



Referencia

- instancia de
- superclase
- singleton class
- * apunta a la clase Class
- ... más singleton classes

Jerarquía de la singleton class de un objeto que no es una clase:

"la *superclase* de la *singleton class* es la *clase*"

Referencia

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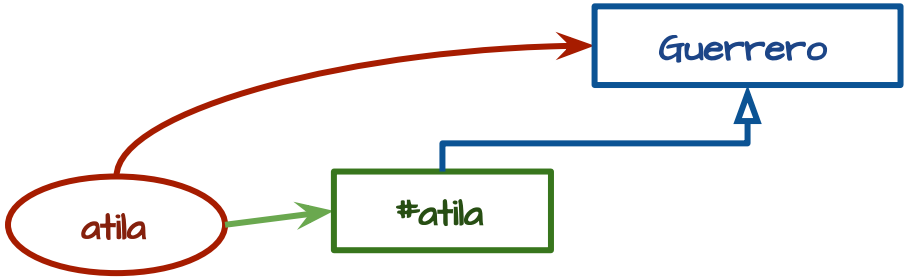
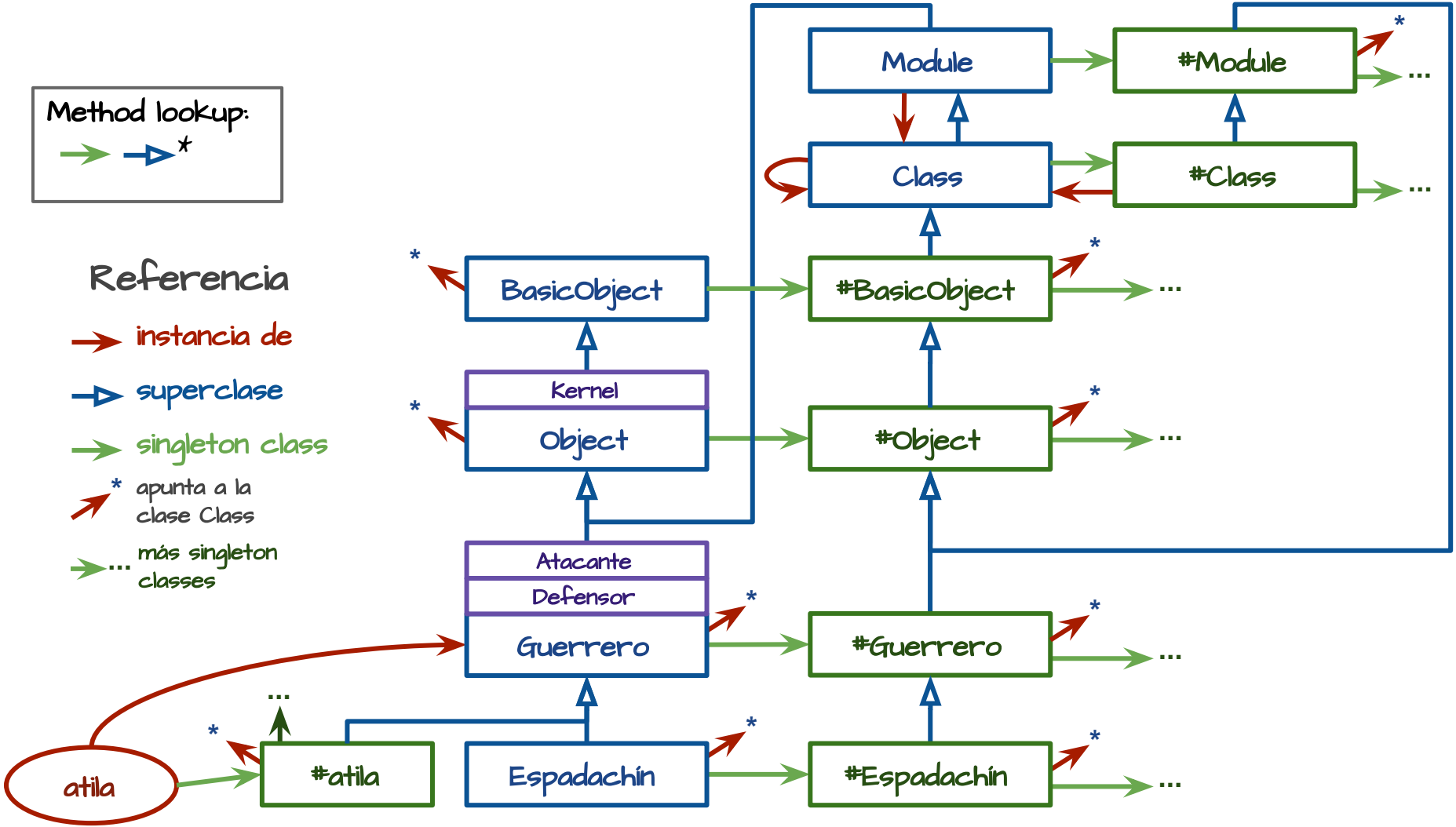
- * apunta a la clase Class
- ... más singleton classes



En resumen

- Todo objeto tiene una singleton class.
  - Esto incluye por supuesto a las clases, y por ende a las singleton classes.
- La superclase de la singleton class de un objeto…
  - Es la clase del objeto (si el objeto no es una clase).
  - Es la singleton class de la superclase del objeto (si el objeto es una clase).
    - Excepto para `BasicObject`.


Method lookup:
*
Referencia
instancia de
superclase
singleton class
* apunta a la clase Class
... más singleton classes
Module
#Module
Class
#Class
BasicObject
#BasicObject
Kernel
Object
#Object
Atacante
Defensor
Guerrero
#Guerrero
atila
#atila
Espadachín
#Espadachín

¿Preguntas?

Intercesión

En donde analizamos las herramientas que nos da Ruby para ejecutar y definir comportamiento dinámicamente.

¿Qué herramientas nos provee Ruby para definir comportamiento?

Clases abiertas

Mensaje *define_method*

(necesitamos pasar un *bloque*)

Comentario: ¿Clases como “moldes”?

Bloques: reificando conjuntos de colaboraciones

Ruby tiene una sintaxis un poco particular...

Instancias de *Proc* (que son objetos) vs. bloques sintácticos (que pueden enviarse como parte de un mensaje)

¿Qué objeto es *self* en el contexto de un bloque?

Familia de métodos *eval/exec*

- `instance_exec`
  Evalúa un bloque en el contexto de un objeto (es decir, `self` dentro del bloque será una referencia al objeto receptor del `instance_exec`).
  Los métodos se definen en la *singleton class* del objeto receptor del mensaje.

- `class_exec/module_exec`
  Evalúa un bloque en el contexto de una clase o un módulo (es decir, `self` dentro del bloque será una referencia al objeto receptor del `class_exec/module_exec`).
  Los métodos se definen en la clase/módulo receptor del mensaje.

¿Preguntas?